



ENGINEERING MATHEMATICS - II

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Department of Science and Humanities

ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 1

Sub Topic : Introduction to Fourier Series

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1	UNIT-V <u>FOURIER SERIES</u>	Introduction to Fourier Series, Dirichlet's conditions, Euler's formula
2		Fourier series of even and odd functions
3 - 4		Problems on Fourier series
5		Half range Fourier series and problems
6		Parseval's identity
7 - 8		Practical harmonic analysis
9 - 10		Complex form of Fourier series

What is Fourier series?

Why do we need Fourier series?

What are the different types of Fourier Series?

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Introduction to Fourier Series

Fourier series is named after the **French Physicist Jean Baptise Fourier (1768 -1830)** who was the first to suggest that periodic signals could be represented by a sum of sinusoids.



What is Fourier Series?

In mathematics, **Fourier series decomposes any periodic function or periodic signal** into the sum of a possibly infinite set of simple oscillating functions namely sines and cosines (or complex exponentials)

Fourier series expansion is used to represent **a periodic signal** in terms of complex exponentials, or equivalently in terms of sine and cosine waveforms.

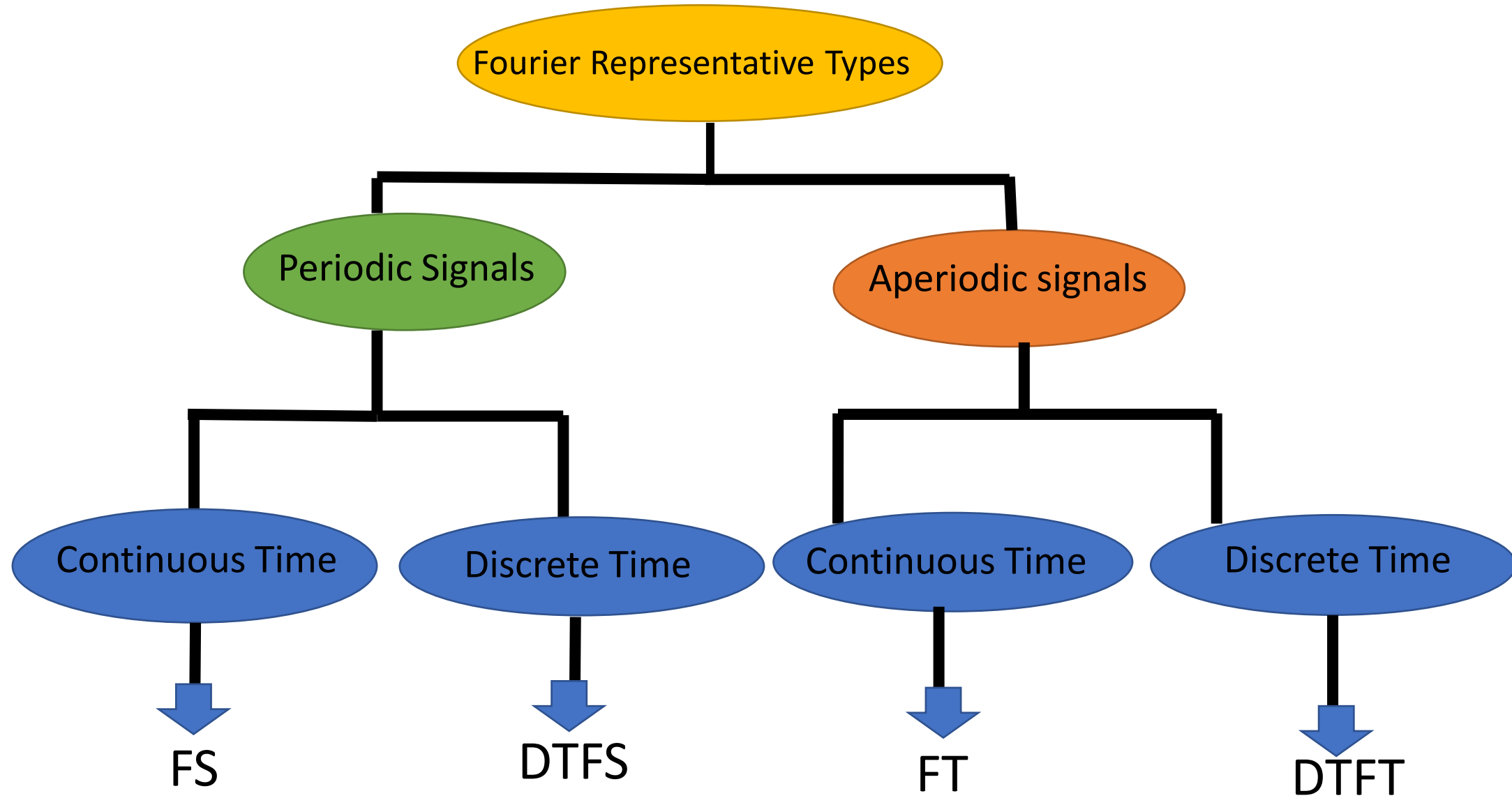
Why do we need Fourier series?

The primary reason that we use Fourier series is that we can better **analyze a signal** in another domain rather in the original domain.

What are the different types of Fourier Series?

The two types of Fourier series are - Trigonometric and exponential. The exponential is more convenient for Fourier series calculations.

Types of Fourier Representation



Applications of Fourier Series in Mathematics



- Fourier series are widely used to solve various ordinary differential equations (ODEs) and partial differential equations (PDEs).
- The Fourier series of functions in the differential equation often gives some prediction about the behaviour of the solution of differential equation. They are useful to find out the dynamics of the solution.

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Applications of Fourier Series in Science and Engineering



- Fourier series has its applications in the field of digital signal processing, wave forms of electrical field, vibrations and conduction of heat etc.
- Also Fourier series finds its applications in electrical engineering, vibration analysis, acoustics, optics, signal processing, image processing, quantum mechanics, econometrics, thin-walled shell theory etc.

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Applications of Fourier Series in Science and Engineering

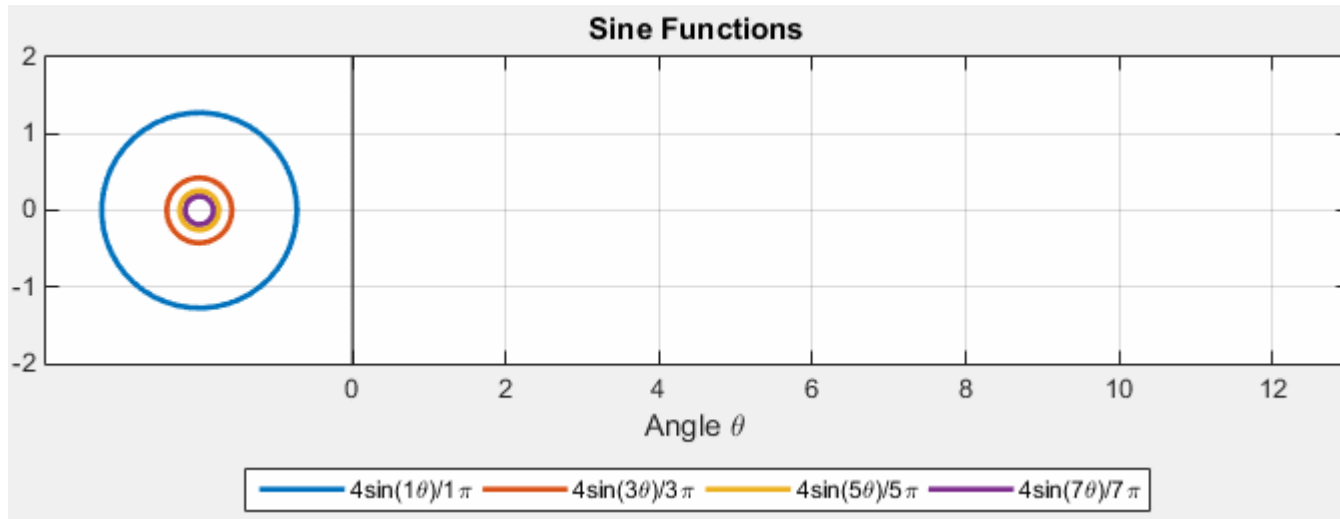
- Fourier series has many applications, infact any field of physical science that uses sinusoidal signals such as engineering, physics and applied mathematics.



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Applications of Fourier Series – Examples

- It can be used to approximate any signal. When we add sines and /or cosines the graph of cosines and sines become closer and closer to the original graph we are trying to approximate.



Applications of Fourier Series - Examples

- **Sound data compression often uses Fourier expansions.**
Take a sound, expand its Fourier series. The compression works by omitting higher terms of that expansion, which will make the signal "smoother", and containing less information to store.

Definition: Periodic function

A function $f(x)$ is said to be periodic if $f(x + T) = f(x)$ for all real x and for some positive number T . Here T is known as the period of the $f(x)$.

❖ Fundamental period of $f(x)$ is the smallest period of f .

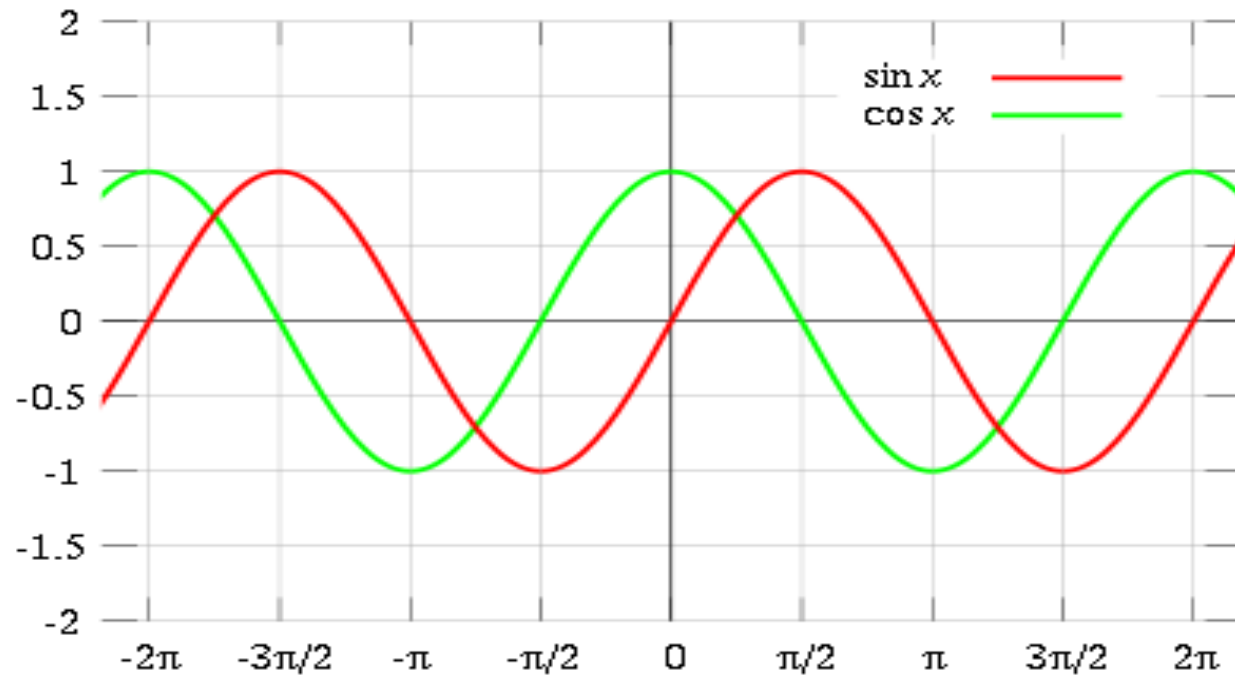
Eg. **$\cos x, \sin x$** ...are periodic functions with period 2π .

Result : If T is the period of $f(x)$, then nT is also period of f for any integer n .

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Fourier Series

Periodic function



Trigonometric Series

Trigonometric series is a fundamental series of the form

$$f(x) = a_0/2 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where constants a_0 , a_n and b_n are called the Euler coefficients.

Euler's (Fourier-Euler) Formulae

Let $f(x)$, a periodic function with period 2π defined in the interval $(c, c + 2\pi)$, be the sum of a trigonometric series,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad (1)$$

$$\text{Then } a_0 = \frac{1}{\pi} \int_c^{c+2\pi} f(x) dx. \quad (2)$$

$$a_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \cos nx dx \text{ for } n = 1, 2, 3 \dots \quad (3)$$

$$b_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \sin nx dx \text{ for } n = 1, 2, 3 \dots \quad (4)$$

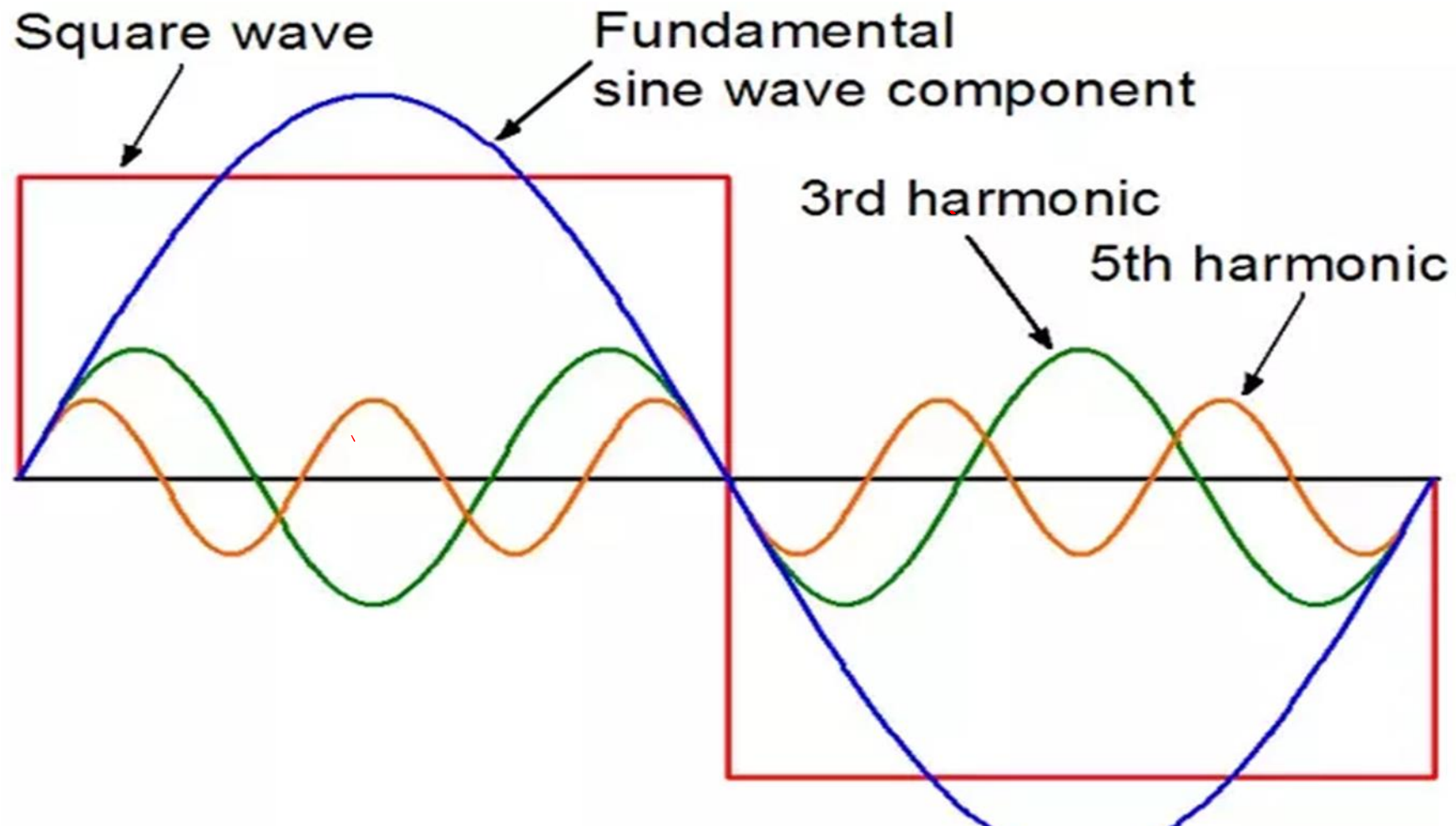
The formula (2),(3) and (4) are known as **Euler formulae** gives the coefficient's a_0, a_n and b_n which are known as **Fourier coefficients of $f(x)$** .

Definition: Fourier Series

Fourier series of a periodic function $f(x)$ with period 2π is the trigonometric series (1) with the Fourier coefficients a_0, a_n and b_n given by the Euler formulae (2),(3) and (4).

The individual terms in Fourier series is known as **harmonics**. The computation and study of Fourier series is known as harmonic analysis.

Harmonics



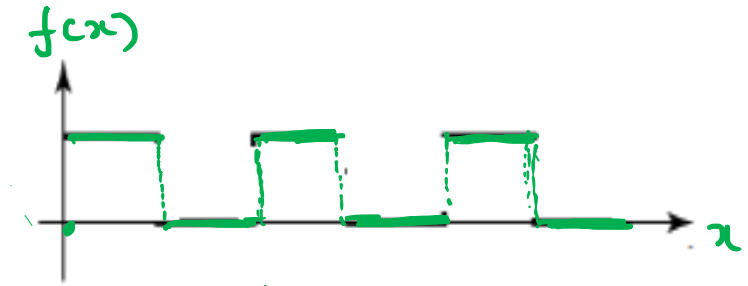
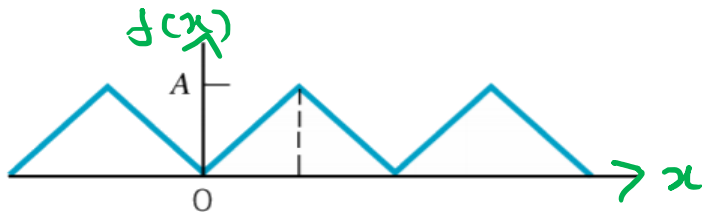
Existence of Fourier Series

Fourier has shown that the expansion of $f(x)$ as an infinite sum of sine and cosine waveforms is possible only if it satisfies certain conditions. These conditions are called Dirichlet conditions which are stated below:

Dirichlet Conditions:

Any function $f(x)$ defined in the interval $c < x < c + 2\pi$ with period 2π is said to satisfy the Dirichlet's conditions if,

- i) $f(x)$ is periodic and single valued function
- ii) $f(x)$ is continuous or $f(x)$ has only a finite number of discontinuities in a period of 2π
- iii) $f(x)$ has a finite number of maxima and minima in a given period.



Note: 1

To find Fourier coefficients a_0 , a_n , b_n when the interval is $(0, 2\pi)$.

If $c = 0$, then the interval of definition for $f(x)$ is $(0, 2\pi)$ and in this case

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \text{ for } n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \text{ for } n = 1, 2, 3 \dots$$

Note: 2

To find Fourier coefficients a_0 , a_n , b_n when the interval is $(-\pi, \pi)$

If $c = -\pi$, then the interval of definition for $f(x)$ is $(-\pi, \pi)$ and in this case

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \text{ for } n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \text{ for } n = 1, 2, 3 \dots$$

Note: 3

At the **point of discontinuity** x_0 , the Fourier series of $f(x)$ converges to the **arithmetic mean** of **left and right hand limit** of $f(x)$ at x_0 .

That is,

$$f(x_0) = \frac{1}{2} [f(x_0 - 0) + f(x_0 + 0)]$$

Some useful results in the computation of Fourier Series

Results:

1. $\cos n\pi = (-1)^n$

2. $\sin n\pi = 0$

3. $\cos \pi = -1$

4. $e^{-\infty} = 0$

5. $\sin \frac{n\pi}{2} = \begin{cases} 0, & n \text{ is even} \\ 1, & n = 1, 5, 9, \dots \\ -1, & n = 3, 7, 11, \dots \end{cases}$

6. $\cos \frac{n\pi}{2} = \begin{cases} 0, & n \text{ is odd} \\ 1, & n = 0, 4, 8, \dots \\ -1, & n = 2, 6, 10, \dots \end{cases}$

Results:

$$7. \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$8. \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

9. Bernoulli's Generalised Formula of Integration by Parts

If u and v are functions of x , then the Bernoulli's rule is

$$\int u * v dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots$$

where $u', u'', u''', \dots$ are successive derivatives of u and $v_1, v_2, v_3, \dots$ are successive integrals of v



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UNIT 5 : Fourier series

Session : 2

Sub Topic : Fourier Series of Arbitrary Period and Fourier Series of Even and Odd Functions

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Recall!!!

We know that, for a periodic function $f(x)$ of period 2π , defined over $(0, 2\pi)$, the Fourier series expansion is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \dots \dots \dots (1)$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \quad \text{for } n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \quad \text{for } n = 1, 2, 3 \dots$$

Recall!!

We know that, for a periodic function $f(x)$ of period 2π , defined over $(-\pi, \pi)$, the Fourier series expansion is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \dots \dots \dots (1)$$

where $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx.$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx, \quad n = 1, 2, 3 \dots$$

Fourier series of functions having arbitrary period $2l$

Fourier series expansion of $f(x)$ on the interval $(-l, l)$

By taking the substitution $y = \frac{\pi x}{l}$ or $x = \frac{ly}{\pi}$

$$\therefore dy = \frac{\pi dx}{l}; \text{ Also, } y = -\pi, x = -l \\ y = \pi, x = l$$

$$F(y) = f\left(\frac{ly}{\pi}\right) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos ny + b_n \sin ny) \text{ where}$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} F(y) dy = \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{ly}{\pi}\right) dy.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(y) \cos ny dy = \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{ly}{\pi}\right) \cos ny dy.$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(y) \sin ny dy = \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{ly}{\pi}\right) \sin ny dy.$$

Fourier series of functions having arbitrary period $2l$

Now, substituting for y in terms of x , we have

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

where

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx.$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx$$

Fourier series of functions having arbitrary period $2l$

The Fourier series expansion of a function $f(x)$ defined over an interval $(-l, l)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

where

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx.$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx$$

Fourier series of functions having arbitrary period $2l$

The Fourier series expansion of a function $f(x)$ defined over an interval $(0, 2l)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

where

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx.$$

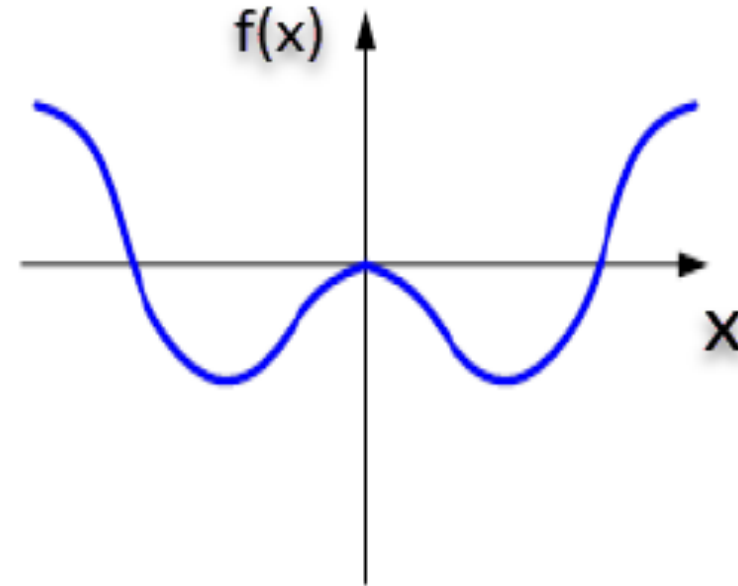
$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

Even Function

A function $f(x)$ is said to be an **even function** if $f(-x) = f(x)$.

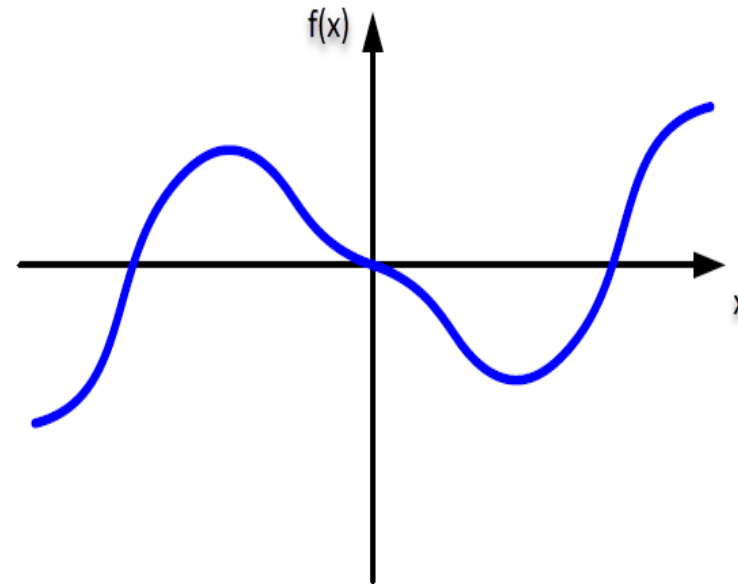
Graphically an even function is **symmetrical about the y-axis**



Odd Function

A function $f(x)$ is said to be an **odd function** if $f(-x) = -f(x)$.

Graphically an odd function is **symmetrical about the origin**.

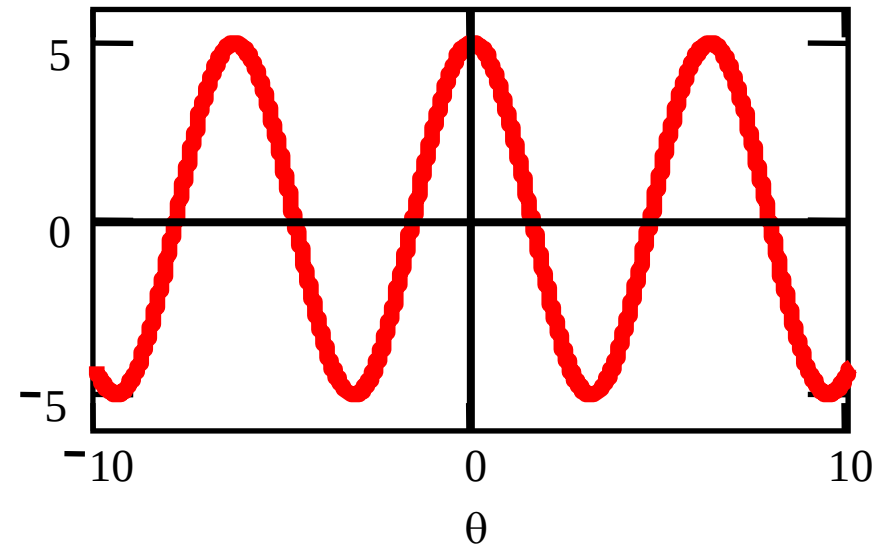


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Fourier Series of Even and Odd Functions

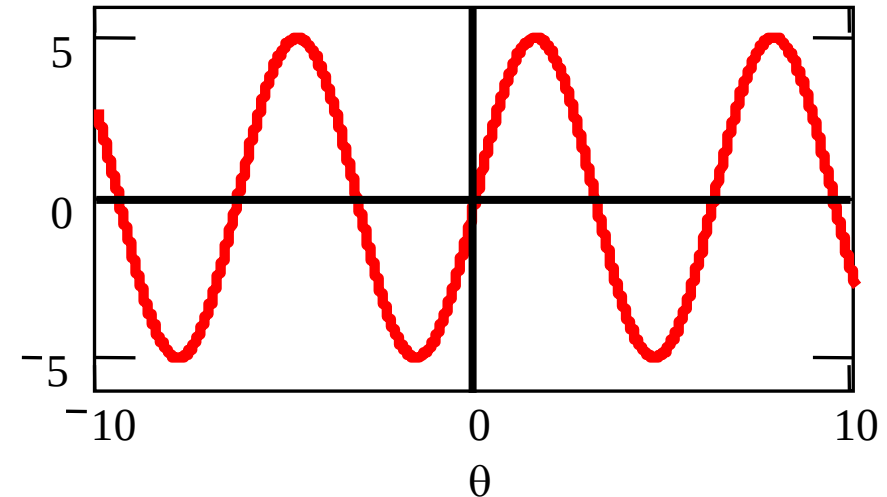
Points to remember

1. Even functions can be represented by cosine waves because, cosine waves are even functions.
2. A sum of even functions is another even function.
3. Product of two even function is even.



Points to remember

1. Odd functions can be represented by sine waves because sine waves are odd functions.
2. A sum of odd functions is another odd function.
3. Product of an odd function and even function is odd.
4. Product of two odd functions is even



Characteristics of Even and Odd Functions

	Even Function	Odd Function
1	A function $f(x)$ is said to be an even function if $f(-x) = f(x)$.	A function $f(x)$ is said to be an odd function if $f(-x) = -f(x)$.
2	Graph of an even function is symmetrical about the y-axis	Graph of an odd function is symmetrical about the x-axis
3	For an even function $\int_{-c}^c f(x)dx = 2 \int_0^c f(x)dx$	For an odd function $\int_{-c}^c f(x)dx = 0$
4	Product of two even functions is even . Product of two odd functions is even .	Product of an even and an odd function is odd .

Fourier Series of an even function in $(-\pi, \pi)$

Case 1:

Suppose $f(x)$ is an **even function** in $(-\pi, \pi)$, then the Fourier series expansion contains only cosine terms and is known as **Fourier cosine series** given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \dots \dots \dots (2)$$

where

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} f(x) dx.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx, n = 1, 2, 3 \dots$$

Fourier Series of an odd function in $(-\pi, \pi)$

Case 2:

Suppose $f(x)$ is an **odd function** in $(-\pi, \pi)$, the Fourier series expansion contains only sine terms and is known as **Fourier sine series** given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx \dots \dots \dots (3)$$

where

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx, n = 1, 2, 3 \dots$$

Fourier Series of an even function defined in $(-l, l)$

Case 1:

Fourier series of an **even function** $f(x)$ in $(-l, l)$ contains only cosine terms and is known as **Fourier cosine series** given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} \right)$$

$$\text{where } a_0 = \frac{1}{l} \int_{-l}^l f(x) dx = \frac{2}{l} \int_0^l f(x) dx$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx, \quad n = 1, 2, 3, \dots$$

Fourier Series of an odd function defined in $(-l, l)$

Case 2:

Fourier series of an **odd function** $f(x)$ in $(-l, l)$ contains only sine terms and is known as **Fourier sine series** given by

$$f(x) = \sum_{n=1}^{\infty} \left(b_n \sin \frac{n\pi x}{l} \right)$$

where

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx, \quad n = 1, 2, 3, \dots$$



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UNIT 5 : Fourier series

Session : 3

Sub Topic : Problems on Fourier Series

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Recall!!

We know that, for a periodic function $f(x)$ of period 2π , defined over $(-\pi, \pi)$, the Fourier series expansion is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \dots \dots \dots (1)$$

where $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx.$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx, \quad n = 1, 2, 3 \dots$$

1. Find the Fourier series of $f(x) = x + x^2$ in $(-\pi, \pi)$. Hence deduce that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Solution:

Given $f(x) = x + x^2$

We have $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \dots \dots (1)$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$
$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x + x^2 dx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx \quad (\text{since } \int_{-\pi}^{\pi} x dx = 0 \text{ is an odd function in the interval } (-\pi, \pi))$$

$$a_0 = \frac{2}{\pi} \left(\frac{x^3}{3} \right)_0^{\pi} = \frac{2\pi^2}{3}$$

Solution contd...

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\&= \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \cos nx dx \\a_n &= \frac{1}{\pi} \left[(x + x^2) \left(\frac{\sin nx}{n} \right) - (1 + 2x) \left(\frac{-\cos nx}{n^2} \right) + (2) \left(\frac{-\sin nx}{n^3} \right) \right]_{-\pi}^{\pi} \\a_n &= \frac{1}{\pi} \left[(1 + 2\pi) \left(\frac{\cos n\pi}{n^2} \right) - (1 - 2\pi) \left(\frac{\cos n\pi}{n^2} \right) \right] \\&= \frac{4}{n^2} \cos n\pi \\a_n &= \frac{4}{n^2} (-1)^n\end{aligned}$$

Solution contd...

$$\begin{aligned}b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \\&= \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \sin nx dx \\&= \left[(x + x^2) \left(\frac{-\cos nx}{n} \right) - (1 + 2x) \left(\frac{-\sin nx}{n^2} \right) + 2 \frac{\cos nx}{n^3} \right]_{-\pi}^{\pi} \\b_n &= \frac{1}{\pi} \left[(\pi + \pi^2) \left(\frac{-\cos n\pi}{n} \right) - (-\pi + \pi^2) \left(\frac{-\cos n\pi}{n} \right) + 2 \frac{\cos n\pi}{n^3} - 2 \frac{\cos n\pi}{n^3} \right] \\&= \frac{-2}{n} \cos n\pi \\b_n &= \frac{-2}{n} (-1)^n\end{aligned}$$

Solution contd...

∴ (1) becomes,

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} (-1)^n \cos nx + \frac{-2}{n} (-1)^n \sin nx \right) \dots\dots\dots (2)$$

This is the required Fourier series

Deduction:

Put $x = \pi$.

Since the function is not defined at π , $f(x)$ at $x = \pi$ is given by

$$\begin{aligned} f(x) &= \frac{(f(\pi+) + f(\pi-))}{2} \\ &= \frac{1}{2} [\pi + \pi^2 - \pi + \pi^2] \end{aligned}$$

Solution contd...

$$f(x) = \pi^2$$

$\therefore$ (2) becomes

$$\pi^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} \right)$$

$$\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \left(\frac{1}{n^2} \right)$$



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ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 4

Sub Topic : Problems on Fourier Series

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**1. Find the Fourier series expansion of $f(x) = x(1 - x)(2 - x)$ in $(0, 2)$.
Deduce the sum of the series $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots$**

Solution:

Given $f(x) = x(1 - x)(2 - x) = x^3 - 3x^2 + 2x$

Here the interval is $(0, 2)$

Comparing with $(0, 2l)$, we have $2l = 2$

$$\therefore l = 1$$

The required Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\pi x + b_n \sin n\pi x)$$

Solution contd...

$$\begin{aligned}a_0 &= \frac{1}{1} \int_0^2 f(x) dx \\&= \int_0^2 x(1-x)(2-x) dx \\&= \int_0^2 (x^3 - 3x^2 + 2x) dx \\&= \left(\frac{x^4}{4} - x^3 + x^2 \right)_0^2 \\a_0 &= 0\end{aligned}$$

Solution contd...

$$\begin{aligned} a_n &= \int_0^2 f(x) \cos \frac{n\pi x}{l} dx \\ &= \int_0^2 x(1-x)(2-x) \cos n\pi x dx \\ &= \int_0^2 (x^3 - 3x^2 + 2x) \cos n\pi x dx \\ &= \left[(x^3 - 3x^2 + 2x) \left(\frac{\sin n\pi x}{n\pi} \right) - (3x^2 - 6x + 2) \left(\frac{-\cos n\pi x}{n^2 \pi^2} \right) \right. \\ &\quad \left. + (6x - 6) \left(\frac{-\sin n\pi x}{n^3 \pi^3} \right) - (6) \left(\frac{\cos n\pi x}{n^4 \pi^4} \right) \right]_0^2 \end{aligned}$$

Solution contd...

$$a_n = \left[\frac{2}{n^2 \pi^2} - \frac{2}{n^2 \pi^2} \right] - 6 \left[\frac{1}{n^4 \pi^4} - \frac{1}{n^4 \pi^4} \right]$$

$$a_n = 0$$

$$\begin{aligned} b_n &= \int_0^2 f(x) \sin \frac{n\pi x}{l} dx \\ &= \int_0^2 x(1-x)(2-x) \sin n\pi x dx \\ &= \int_0^2 (x^3 - 3x^2 + 2x) \sin n\pi x dx \end{aligned}$$

Solution contd..

$$\begin{aligned} b_n &= \left[(x^3 - 3x^2 + 2x) \left(\frac{-\cos n\pi x}{n\pi} \right) - (3x^2 - 6x + 2) \left(\frac{-\sin n\pi x}{n^2\pi^2} \right) \right. \\ &\quad \left. + (6x - 6) \left(\frac{\cos n\pi x}{n^3\pi^3} \right) - (6) \left(\frac{\sin n\pi x}{n^4\pi^4} \right) \right]_0^2 \\ b_n &= [0 - 0] + 6 \left[\frac{1}{n^3\pi^3} + \frac{1}{n^3\pi^3} \right] \end{aligned}$$

$$b_n = \frac{12}{n^3\pi^3}$$

Solution contd...

Hence the required Fourier series is

$$f(x) = x(1 - x)(2 - x) = \sum_{n=1}^{\infty} \left(\frac{12}{n^3 \pi^3} \sin n\pi x \right) \dots\dots(1)$$

Deduction

Put $x = \frac{1}{2}$ in (1)

$$1 - \frac{3}{4} + \frac{1}{8} = \sum_{n=1}^{\infty} \left(\frac{12}{n^3 \pi^3} \sin \frac{n\pi}{2} \right)$$
$$\frac{3}{8 * 12} = \frac{1}{\pi^3} \left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots \right]$$
$$\frac{\pi^3}{32} = \left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots \right]$$

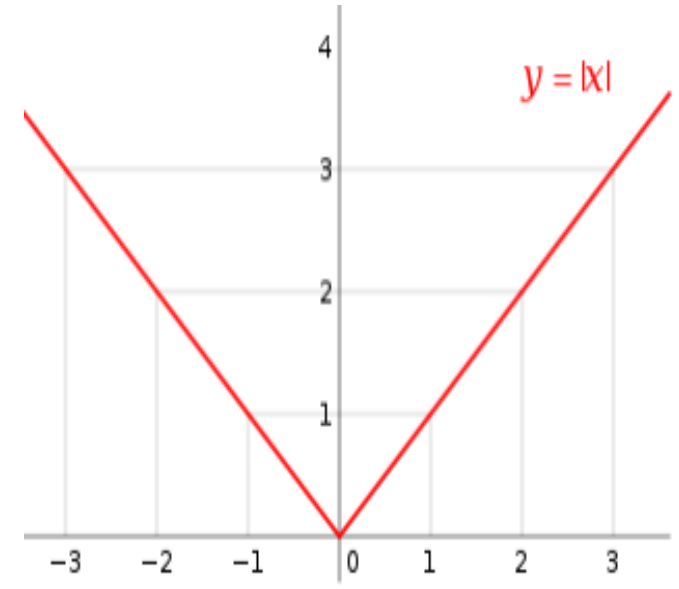
2. Obtain the Fourier Series to represent the function $f(x) = |x|$, $-\pi < x < \pi$.

Solution:

Since $f(x)$ is an even function, the Fourier Series expansion of $f(x)$ is given by

$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$ where

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} x dx \\ &= \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} \end{aligned}$$



$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Solution Contd....

$$\therefore a_0 = \frac{\pi}{2}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx$$

$$= \frac{2}{\pi} \left[x \cdot \frac{(\sin nx)}{n} - (1) \cdot \frac{(-\cos nx)}{n^2} \right]_0^{\pi}$$

$$\therefore a_n = \frac{2}{\pi n^2} [(-1)^n - 1]$$

Therefore the required Fourier Series is given by

$$f(x) = |x| = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$|x| = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} [(-1)^n - 1] \cos nx$$



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ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 5

Sub Topic : Half Range Fourier Series

Department of Science and Humanities

Half Range Fourier series

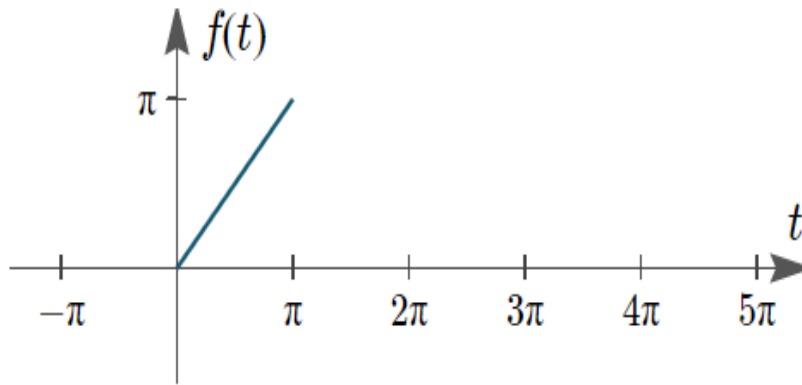
- ❖ Consider a function $f(x)$ for which the Fourier series is required is defined over half the range from 0 to π (or 0 to l).
- ❖ In such cases, the given function can be extended to the other half range, as either an odd function or an even function.
- ❖ The Fourier expansion of such a function will consist either only sine or only cosine terms.

The series produced is then called a **half range Fourier series**.

Half Range Fourier series

Example:

Consider the function $f(t) = t, 0 < t < \pi$ to be half of a period π of a 2π periodic function. The extension can be done to the other half either as an odd function or as an even function.

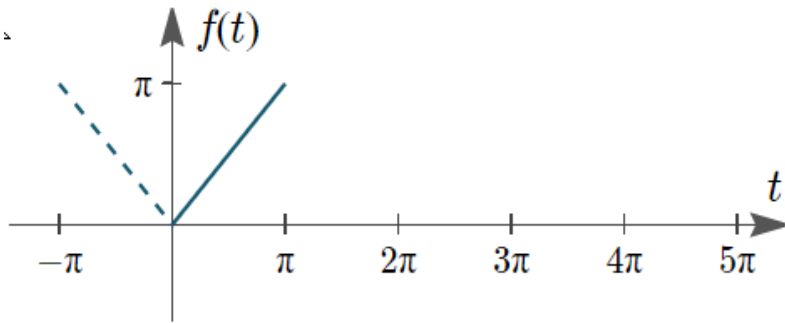


Graph of $f(t)$.

Half Range Fourier series

Case (i) Even Extension

We will extend this function to the other half of the range $(-\pi, 0)$ as an even function.



Graph of $f(t)$, illustrating it's an even function.

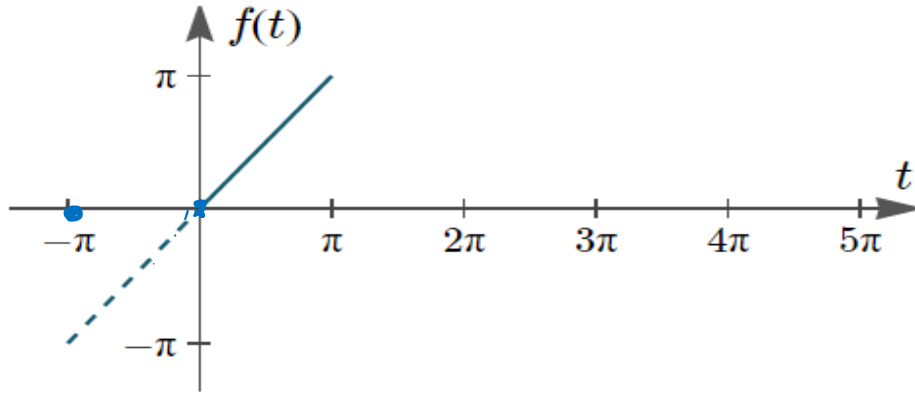
So we have an even function in the interval $(-\pi, \pi)$ and the fourier series expansion will consists of only **cosine terms** and is called **Half Range Fourier Cosine Series**.

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Half Range Fourier series

Case (ii) Odd Extension

We will extend this function to the other half of the range $(-\pi, 0)$ as an odd function.



Graph of $f(t)$, showing it's an odd function.

So we have an odd function in the interval $(-\pi, \pi)$ and the Fourier series expansion will consist of only **sine terms** and is called **Half Range Fourier Sine Series**.

Half Range Cosine series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

Fourier Coefficients:

$$\diamond a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx.$$

$$\diamond a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

Half Range Sine Series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

Fourier Coefficients:

$$\diamond b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

Half Range Fourier series of $f(x)$ in $(0, l)$

Half Range Cosine series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}$$

Fourier Coefficients:

$$\diamond a_0 = \frac{2}{l} \int_0^l f(x) dx.$$

$$\diamond a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$$

Half Range Sine Series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

Fourier Coefficients:

$$\diamond b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

1. Find the Fourier half range a) Cosine series b) Sine series of

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2 \end{cases}$$

Solution:

a) Cosine series: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}$

Comparing (0,2) with half the range (0,l), we have $l = 2$

Therefore $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{2}$

$$\begin{aligned} a_0 &= \frac{2}{2} \int_0^2 f(x) dx \\ &= \int_0^1 x dx + \int_1^2 (2 - x) dx \end{aligned}$$

$$a_0 = \left[\frac{x^2}{2} \right]_0^1 + \left[2x - \frac{x^2}{2} \right]_1^2$$

$$a_0 = 1$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$$

$$a_n = \frac{2}{2} \int_0^2 f(x) \cos \frac{n\pi x}{2} dx$$

$$= \int_0^1 x \cos \frac{n\pi x}{2} dx + \int_1^2 (2-x) \cos \frac{n\pi x}{2} dx$$

$$= \left[x \cdot \frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right]_0^1 - \left[-\frac{\cos \frac{n\pi x}{2}}{\frac{n^2 \pi^2}{4}} \right]_0^1 + \left[(2-x) \cdot \frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right]_1^2 - \left[\frac{\cos \frac{n\pi x}{2}}{\frac{n^2 \pi^2}{4}} \right]_1^2$$

$$= \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) + \frac{4}{n^2\pi^2} \left[\cos\left(\frac{n\pi}{2}\right) - 1 \right] - \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2} \left[\cos n\pi - \cos\left(\frac{n\pi}{2}\right) \right]$$

$$a_n = \frac{8}{n^2\pi^2} \cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2} [1 + (-1)^n]$$

$\therefore$ Cosine series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{2}$$

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} \cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2} [1 + (-1)^n] \cos \frac{n\pi x}{2}$$

Problems on Half Range Fourier series

b) Sine series : $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$

where $b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$

Comparing (0,2) with half the range (0,l), we have $l = 2$

$$\therefore f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{2}$$

$$b_n = \frac{2}{2} \int_0^2 f(x) \sin \frac{n\pi x}{2} dx$$

$$= \int_0^1 x \sin \frac{n\pi x}{2} dx + \int_1^2 (2-x) \sin \frac{n\pi x}{2} dx$$

$$= \left[x \cdot \frac{-\cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right]_0^1 - \left[-\frac{\sin \frac{n\pi x}{2}}{\frac{n^2 \pi^2}{4}} \right]_0^1 + \left[(2-x) \cdot \frac{-\cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right]_1^2 - \left[\frac{\sin \frac{n\pi x}{2}}{\frac{n^2 \pi^2}{4}} \right]_1^2$$

$$b_n = \frac{-2}{n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{4}{n^2\pi^2} \left[\sin\left(\frac{n\pi}{2}\right) \right] + \frac{2}{n\pi} \cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2} \left[0 - \sin\left(\frac{n\pi}{2}\right) \right]$$

$$b_n = \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)$$

∴ Sine series is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

$$f(x) = \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \sin \frac{n\pi x}{2}$$



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ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 6

Sub Topic : Parseval's Identity

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Parseval's Identity

Parseval's identity relates the squares of the Fourier coefficients to the integral of the square of the function $f(x)$ of period $2l$ defined in the interval $(-l, l)$. That is, if

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right) \dots \dots (1)$$

then,

$$\int_{-l}^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$$

Proof:

Multiplying both sides of (1) by $f(x)$ and integrating term by term from $-l$ to l we get

$$\int_{-l}^l [f(x)]^2 dx$$
$$= \frac{a_0}{2} \int_{-l}^l f(x) dx + \sum_{n=1}^{\infty} \left(a_n \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx + b_n \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx \right) \dots (2)$$

Now,

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx \Rightarrow \int_{-l}^l f(x) dx = l a_0$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx \Rightarrow \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = l a_n$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx \Rightarrow \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = l b_n$$

Parseval's Identity

Therefore (2) becomes

$$\int_{-l}^l [f(x)]^2 dx = \frac{a_0}{2} (la_0) + \sum_{n=1}^{\infty} (a_n(la_n) + b_n(lb_n))$$

$$\text{Hence } \int_{-l}^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$$

which is the desired Parseval's formula

Parseval's Identity

Sl.No	Period of $f(x)$	Parseval's Identity
1.	$(-l, l)$	$\int_{-l}^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$
2.	$(0, 2l)$	$\int_0^{2l} [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$

ENGINEERING MATHEMATICS -II

Parseval's Identity of Even and Odd Functions



Sl.No	Period of $f(x)$	Parseval's Identity
3.	$(-l, l)$	$2\int_0^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2 \right]$ (Even Function)
4.	$(-l, l)$	$2\int_0^l [f(x)]^2 dx = l \left[\sum_{n=1}^{\infty} b_n^2 \right]$ (Odd function)

Sl.No	Period of $f(x)$	Parseval's Identity
3.	$(0, l)$	$\underline{2 \int_0^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2 \right]}$ (Half range cosine series)
4.	$(0, l)$	$2 \int_0^l [f(x)]^2 dx = l \left[\sum_{n=1}^{\infty} b_n^2 \right]$ (Half range sine series)

Obtain the Fourier Series for the function $f(x) = x^2, -\pi < x < \pi$. Hence show that $\sum \frac{1}{n^4} = \frac{\pi^4}{90}$.

Solution:

Here $f(x) = f(-x)$. So x^2 is an even function.

Hence we have

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx, \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx, \quad b_n = 0$$

and the Fourier expansion of $f(x)$ over $(-\pi, \pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

Parseval's Identity - Problems

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx,$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx,$$

$$= \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi}$$

$$a_0 = \frac{2\pi^2}{3}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} (x^2) \cos nx \, dx$$

$$= \frac{2}{\pi} \left[(x^2) \left(\frac{\sin nx}{n} \right) - (2x) \left(-\frac{\cos nx}{n^2} \right) + 2 \left(-\frac{\sin nx}{n^3} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[(2x) \left(\frac{\cos nx}{n^2} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi n^2} [(2\pi) (\cos n\pi - 0)]$$

$$= \frac{1}{\pi n^2} \cos n\pi (4\pi) =$$

$$a_n = \frac{4}{n^2} (-1)^n$$

Parseval's Identity - Problems

Therefore the required Fourier series is given by,

$$x^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} (-1)^n \cos nx \right)$$

To deduce, we will use Parseval's identity for even functions.

$$2 \int_0^{\pi} [f(x)]^2 dx = \pi \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2 \right]$$

$$\text{Therefore, } 2 \int_0^{\pi} [x^2]^2 dx = \pi \left[\frac{1}{2} \frac{4\pi^4}{9} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} (-1)^n \right)^2 \right]$$

$$2 \left[\frac{x^5}{5} \right]_0^{\pi} = \frac{4\pi^5}{18} + 16\pi \sum_{n=1}^{\infty} \frac{1}{n^4}$$

$$2 \frac{\pi^5}{5} - \frac{2\pi^5}{9} = 16\pi \sum_{n=1}^{\infty} \frac{1}{n^4}$$

$$\Rightarrow \sum \frac{1}{n^4} = \frac{\pi^4}{90}$$



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ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 7

Sub Topic : Practical Harmonic Analysis

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ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis

In many engineering and scientific problems, $f(x)$ is not given directly, rather set of discrete values of function are given in the form (x_i, y_i) , $i = 1, 2, 3, \dots, m$ where x_i 's are equispaced. The process of obtaining $f(x)$ in terms of Fourier series from given set of values (x_i, y_i) , is known as practical harmonic analysis.

x	x_1	x_2	x_3				x_N
$y = f(x)$	y_1	y_2	y_3				y_N

Recall!!!

We know that, for a periodic function $f(x)$ of period 2π , defined over $(0, 2\pi)$, the Fourier series expansion is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \dots \dots \dots (1)$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \quad \text{for } n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \quad \text{for } n = 1, 2, 3 \dots$$

Equation (1) can be rearranged as

$$f(x) = \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x) + (a_3 \cos 3x + b_3 \sin 3x) + \dots$$

The term $(a_1 \cos x + b_1 \sin x)$ is called the **First Harmonic**.

The term $(a_2 \cos 2x + b_2 \sin 2x)$ is called the **Second Harmonic**.

The term $(a_3 \cos 3x + b_3 \sin 3x)$ is called the **Third Harmonic** and so on.

Fourier series of functions having arbitrary period $2l$

Recall!!!

The Fourier series expansion of a function $f(x)$ defined over an interval $(0, 2l)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right) \dots \dots \dots (2)$$

where

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx.$$

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

Equation (2) can be rearranged as

$$f(x) = \frac{a_0}{2} + (a_1 \cos \frac{\pi x}{l} + b_1 \sin \frac{\pi x}{l}) + (a_2 \cos \frac{2\pi x}{l} + b_2 \sin \frac{2\pi x}{l}) + (a_3 \cos \frac{3\pi x}{l} + b_3 \sin \frac{3\pi x}{l}) + \dots$$

The term $(a_1 \cos \frac{\pi x}{l} + b_1 \sin \frac{\pi x}{l})$ is called the **First Harmonic**.

The term $(a_2 \cos \frac{2\pi x}{l} + b_2 \sin \frac{2\pi x}{l})$ is called the **Second Harmonic**.

The term $(a_3 \cos \frac{3\pi x}{l} + b_3 \sin \frac{3\pi x}{l})$ is called the **Third Harmonic** and so on.

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis



Suppose we have a set of N values of $y = f(x)$ having period 2π or $2l$ at equidistant points of x in the interval $(0, 2\pi)$ or $(0, 2l)$ then Fourier coefficients a_0 , a_n and b_n are calculated using the relations given below:

Period	Fourier Coefficients		
2π	$a_0 = \frac{2}{N} \sum y$	$a_n = \frac{2}{N} \sum y \cos nx$	$b_n = \frac{2}{N} \sum y \sin nx$
$2l$	$a_0 = \frac{2}{N} \sum y$	$a_n = \frac{2}{N} \sum y \cos\left(\frac{n\pi x}{l}\right)$	$b_n = \frac{2}{N} \sum y \sin\left(\frac{n\pi x}{l}\right)$

Note:

1. Here **N** denotes the **number of observations**.
2. The amplitude of the first harmonic is given by $\sqrt{a_1^2 + b_1^2}$.
Similarly the amplitudes of the second and third harmonics are given by $\sqrt{a_2^2 + b_2^2}$ and $\sqrt{a_3^2 + b_3^2}$ respectively.

Example:

The following values of 'y' give the displacement of a machine part for the rotation 'x' of a flywheel. Express y as a Fourier series up to first harmonics.

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
y	7.9	7.2	3.6	0.5	0.9	11.7	7.9

Solution:

Here, length of interval is 2π .

Here $N=6$ since $[y]_{0^\circ} = [y]_{360^\circ}$.

Fourier expansion upto the first harmonic is given by

$$f(x) = \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x)$$

$$a_0 = \frac{2}{N} \sum y$$

$$= \frac{2}{6} \sum y = \frac{1}{3} (26.9) = 8.9667$$

$$a_n = \frac{2}{N} \sum y \cos nx$$

$$\begin{aligned} a_1 &= \frac{2}{6} \sum y \cos x \\ &= \frac{1}{3} \sum y \cos x \end{aligned}$$

$$b_n = \frac{2}{N} \sum y \sin nx$$

$$\begin{aligned} b_1 &= \frac{2}{6} \sum y \sin x \\ &= \frac{1}{3} \sum y \sin x \end{aligned}$$

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis - Problems



x	y	$\cos x$	$y \cos x$	$\sin x$	$y \sin x$
0	7.9	1	7.9	0	0
$\pi/3$	7.2	0.5	3.6	0.866	6.2352
$2\pi/3$	3.6	-0.5	-1.8	0.866	3.1176
π	0.5	-1	-0.5	0	0
$4\pi/3$	0.9	-0.5	-0.45	-0.866	-0.7794
$5\pi/3$	6.8	0.5	3.4	-0.866	-5.8888
Total	26.9		12.15		2.6846

$$a_0 = \frac{2}{N} \sum y$$
$$= \frac{2}{6} \sum y = \frac{1}{3} (26.9) = 8.9667$$

$$a_n = \frac{2}{N} \sum y \cos nx$$

$$a_1 = \frac{2}{6} \sum y \cos x$$
$$= \frac{1}{3} \sum y \cos x$$
$$= \frac{1}{3} [12.15] = 4.05$$

$$b_n = \frac{2}{N} \sum y \sin nx$$

$$b_1 = \frac{2}{6} \sum y \sin x$$
$$= \frac{1}{3} \sum y \sin x$$
$$= \frac{1}{3} [2.6846] = 0.8949$$

Therefore Fourier series up to first harmonic is given by

$$f(x) = \frac{a_0}{2} + a_1 \cos x + b_1 \sin x$$

$$y = 4.48335 + 4.05 \cos x + 0.8949 \sin x$$



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ENGINEERING MATHEMATICS - II

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ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 8

Sub Topic : Problems on Practical Harmonic Analysis

Department of Science and Humanities

Suppose we have a set of N values of $y = f(x)$ having period 2π or $2l$ at equidistant points of x in the interval $(0, 2\pi)$ or $(0, 2l)$ then Fourier coefficients a_0 , a_n and b_n are calculated using the relations given below:

Period	Fourier Coefficients		
2π	$a_0 = \frac{2}{N} \sum y$	$a_n = \frac{2}{N} \sum y \cos nx$	$b_n = \frac{2}{N} \sum y \sin nx$
$2l$	$a_0 = \frac{2}{N} \sum y$	$a_n = \frac{2}{N} \sum y \cos\left(\frac{n\pi x}{l}\right)$	$b_n = \frac{2}{N} \sum y \sin\left(\frac{n\pi x}{l}\right)$

Note:

1. Here **N** denotes the **number of observations**.
2. The amplitude of the first harmonic is given by $\sqrt{a_1^2 + b_1^2}$.
Similarly the amplitudes of the second and third harmonics are given by $\sqrt{a_2^2 + b_2^2}$ and $\sqrt{a_3^2 + b_3^2}$ respectively.

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis - Problems



1. Determine the constant term and the first two coefficients of cosine and sine terms in the Fourier series for the following data.

x	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$
y	0	9.2	14.4	17.8	17.3	11.7

Solution:

The interval of x is $0 \leq x < \pi$.

Here, length of interval is π . That is, $l = \frac{\pi}{2}$

Here $N=6$.

$$f(x) = \frac{a_0}{2} + (a_1 \cos \frac{\pi x}{l} + b_1 \sin \frac{\pi x}{l}) + (a_2 \cos \frac{2\pi x}{l} + b_2 \sin \frac{2\pi x}{l}) + \dots$$

Substituting for $l = \frac{\pi}{2}$

$$f(x) = \frac{a_0}{2} + (a_1 \cos 2x + b_1 \sin 2x) + (a_2 \cos 4x + b_2 \sin 4x) + \dots$$

$$\begin{aligned}a_0 &= \frac{2}{N} \sum y \\ &= \frac{2}{6} \sum y = \frac{1}{3} (70.4) = 23.46\end{aligned}$$

$$a_n = \frac{2}{N} \sum y \cos \left(\frac{n\pi x}{l} \right)$$

$$= \frac{2}{6} \sum y \cos \left(\frac{n\pi x}{\pi/2} \right)$$

$$a_n = \frac{1}{3} \sum y \cos (2nx)$$

Put $n=1,2$

$$a_1 = \frac{1}{3} \sum y \cos 2x$$

$$a_2 = \frac{1}{3} \sum y \cos 4x$$

$$b_n = \frac{2}{N} \sum y \sin \left(\frac{n\pi x}{l} \right)$$

$$= \frac{2}{6} \sum y \sin \left(\frac{n\pi x}{\pi/2} \right)$$

$$b_n = \frac{1}{3} \sum y \sin (2nx)$$

$$b_1 = \frac{1}{3} \sum y \sin 2x$$

$$b_2 = \frac{1}{3} \sum y \sin 4x$$

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis - Problems

x	y	$\cos 2x$	$y \cos 2x$	$\cos 4x$	$y \cos 4x$
0	0	1	0	1	0
$\pi/6$	9.2	0.5	4.6	-0.5	-4.6
$\pi/3$	14.4	-0.5	-7.2	-0.5	-7.2
$\pi/2$	17.8	-1	-17.8	1	17.8
$2\pi/3$	17.3	-0.5	-8.65	-0.5	-8.65
$5\pi/6$	11.7	0.5	5.85	-0.5	-5.85
Total			-23.2		-8.5

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis - Problems



$$\begin{aligned}a_1 &= \frac{1}{3} \sum y \cos 2x \\ &= \frac{1}{3} [-23.2] = -7.73\end{aligned}$$

$$\begin{aligned}a_2 &= \frac{1}{3} \sum y \cos 4x \\ &= \frac{1}{3} [-8.5] = -2.83\end{aligned}$$

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis - Problems

x	y	$\sin 2x$	$y \sin 2x$	$\sin 4x$	$y \sin 4x$
0	0	0	0	0	0
$\pi/6$	9.2	0.866	7.9672	0.866	7.9672
$\pi/3$	14.4	0.866	12.4704	-0.866	-12.4704
$\pi/2$	17.8	0	0	0	0
$2\pi/3$	17.3	-0.866	-14.9818	0.866	14.9818
$5\pi/6$	11.7	-0.866	-10.1322	-0.866	-10.1322
Total			-4.6764		0.3424

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis - Problems

$$\begin{aligned} b_1 &= \frac{1}{3} \sum y \sin 2x \\ &= \frac{1}{3} [-4.6764] = -1.2124 \end{aligned}$$

$$\begin{aligned} b_2 &= \frac{1}{3} \sum y \sin 4x \\ &= \frac{1}{3} [0.3464] = 0.1155 \end{aligned}$$

Therefore Fourier series is given by

$$f(x) = \frac{a_0}{2} + a_1 \cos 2x + b_1 \sin 2x + a_2 \cos 4x + b_2 \sin 4x$$

$$y = 11.733 - 7.733 \cos 2x - 1.2124 \sin 2x - 2.833 \cos 4x + 0.116 \sin 4x$$

Practical Harmonic Analysis - Problems

2. The following table gives the variations of periodic current over a period T .

t (sec)	0	$T/6$	$T/3$	$T/2$	$2T/3$	$5T/6$	T
A (amp)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98

Show that there is a direct current part 0.75 amp in the variable current and obtain the first two harmonics of the Fourier series.

Solution:

The interval of t is $0 \leq t < T$

Here, length of interval is T. That is, $l = \frac{T}{2}$

We observe that the values of A at $t = 0$ and $t = T$ are the same. Hence we omit the last value.

Here **N= 6**

Fourier series up to the second harmonics is given by

$$f(x) = \frac{a_0}{2} + \left(a_1 \cos \frac{\pi x}{l} + b_1 \sin \frac{\pi x}{l}\right) + \left(a_2 \cos \frac{2\pi x}{l} + b_2 \sin \frac{2\pi x}{l}\right)$$

Substituting for $l = \frac{T}{2}$

$$f(x) = \frac{a_0}{2} + \left(a_1 \cos \frac{2\pi x}{T} + b_1 \sin \frac{2\pi x}{T}\right) + \left(a_2 \cos \frac{4\pi x}{T} + b_2 \sin \frac{4\pi x}{T}\right)$$

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis - Problems

$$\begin{aligned}a_0 &= \frac{2}{N} \sum y \\&= \frac{2}{6} \sum y \\&= \frac{1}{3} (4.5) \\a_0 &= \mathbf{1.5}\end{aligned}$$



$$a_n = \frac{2}{N} \sum y \cos\left(\frac{n\pi x}{l}\right)$$

$$= \frac{2}{6} \sum y \cos\left(\frac{n\pi x}{T/2}\right)$$

$$a_n = \frac{1}{3} \sum y \cos\left(\frac{2n\pi x}{T}\right)$$

$$b_n = \frac{2}{N} \sum y \sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{2}{6} \sum y \sin\left(\frac{n\pi x}{T/2}\right)$$

$$= \frac{1}{3} \sum y \sin\left(\frac{2n\pi x}{T}\right)$$

Put n=1,2

$$a_1 = \frac{1}{3} \sum y \cos\left(\frac{2\pi x}{T}\right)$$

$$a_2 = \frac{1}{3} \sum y \cos\left(\frac{4\pi x}{T}\right)$$

$$b_1 = \frac{1}{3} \sum y \sin\left(\frac{2\pi x}{T}\right)$$

$$b_2 = \frac{1}{3} \sum y \sin\left(\frac{4\pi x}{T}\right)$$

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Practical Harmonic Analysis - Problems



$x(t)$	$y(A)$	$\cos\left(\frac{2\pi x}{T}\right)$	$y \cos\left(\frac{2\pi x}{T}\right)$	$\cos\left(\frac{4\pi x}{T}\right)$	$y \cos\left(\frac{4\pi x}{T}\right)$
0	1.98	1	1.98	1	1.98
$T/6$	1.30	0.5	0.65	-0.5	-0.65
$T/3$	1.05	-0.5	-0.525	-0.5	-0.525
$T/2$	1.30	-1	-1.3	1	1.3
$2T/3$	-0.88	-0.5	0.44	-0.5	0.44
$5T/6$	-0.25	0.5	-0.125	-0.5	0.125
Total			1.12		2.67

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis - Problems

$x(t)$	$y(A)$	$\sin\left(\frac{2\pi x}{T}\right)$	$y \sin\left(\frac{2\pi x}{T}\right)$	$\sin\left(\frac{4\pi x}{T}\right)$	$y \sin\left(\frac{4\pi x}{T}\right)$
0	1.98	0	0	0	0
$T/6$	1.30	0.866	1.1258	0.866	1.1258
$T/3$	1.05	0.866	0.9093	-0.866	-0.9093
$T/2$	1.30	0	0	0	0
$2T/3$	-0.88	-0.866	0.76208	0.866	-0.76208
$5T/6$	-0.25	-0.866	0.2165	-0.866	0.2165
Total			3.01368		-0.32908

$$\frac{a_0}{2} = \frac{1.5}{2} = 0.75$$

Therefore, the direct current part = 0.75 amperes

$$a_1 = \frac{1}{3} \sum y \cos \left(\frac{2\pi x}{T} \right) = \frac{1}{3} [1.12] = 0.3733$$

$$b_1 = \frac{1}{3} \sum y \sin \left(\frac{2\pi x}{T} \right) = \frac{1}{3} [3.01368] = 1.00456$$

$$a_2 = \frac{1}{3} \sum y \cos \left(\frac{4\pi x}{T} \right) = \frac{1}{3} [2.67] = 0.89$$

$$b_2 = \frac{1}{3} \sum y \sin \left(\frac{2\pi x}{T} \right) = \frac{1}{3} [-0.32908] = -0.1097$$

Fourier series up to the second harmonics is given by

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \left(a_1 \cos \frac{2\pi x}{T} + b_1 \sin \frac{2\pi x}{T} \right) + \left(a_2 \cos \frac{4\pi x}{T} + b_2 \sin \frac{4\pi x}{T} \right) \\ &= 0.75 + \left(0.3733 \cos \frac{2\pi x}{T} + 1.00456 \sin \frac{2\pi x}{T} \right) + \left(0.89 \cos \frac{4\pi x}{T} - \right. \\ &\quad \left. 0.1097 \sin \frac{4\pi x}{T} \right) \end{aligned}$$



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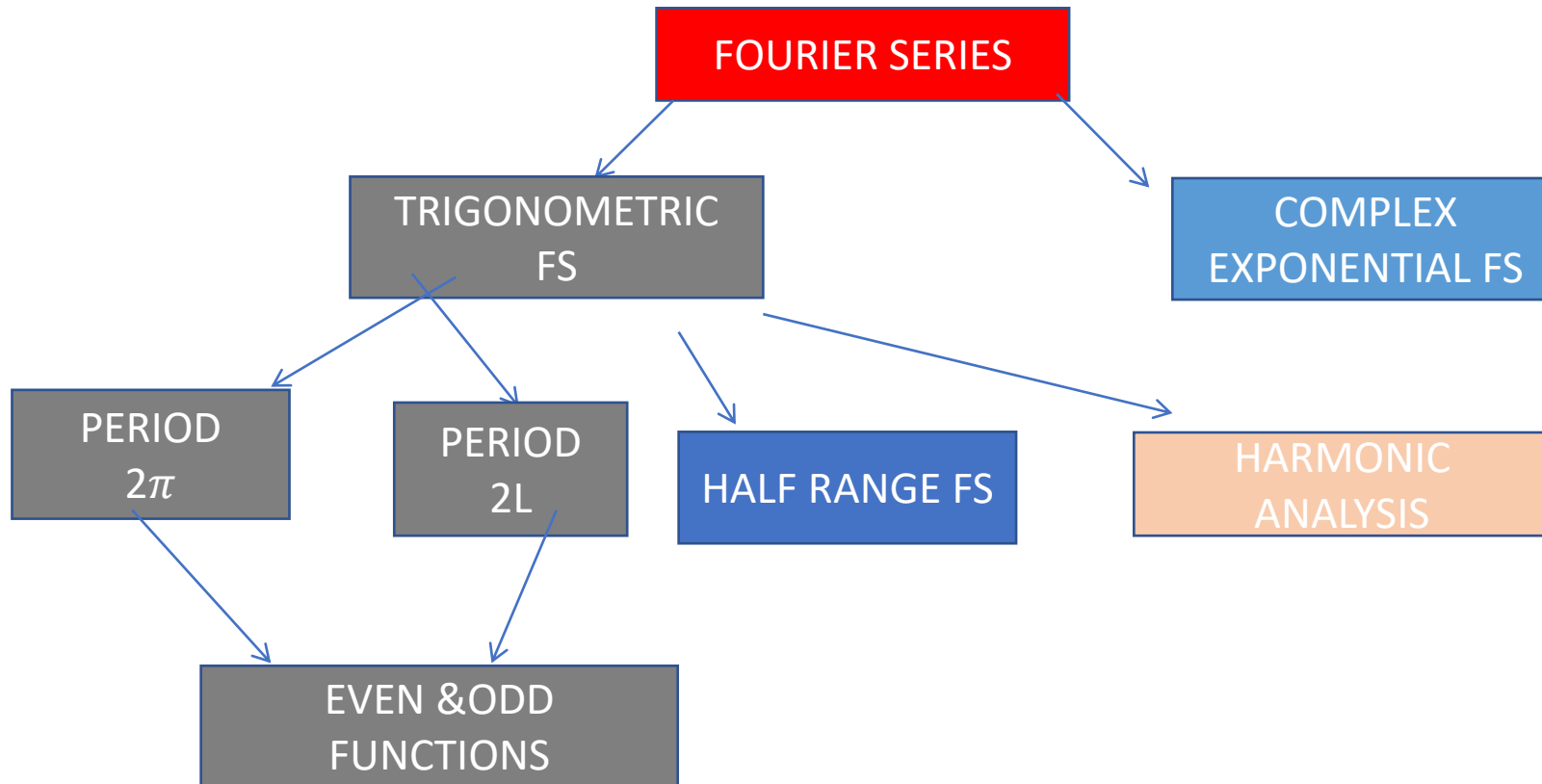
ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 9

Sub Topic : Complex Form of the Fourier Series

Department of Science and Humanities



Complex form of the Fourier series



- ❖ The representation of signals over a certain time interval in terms of linear composition of exponential functions is called exponential Fourier series or complex form of the Fourier series.
- ❖ The complex form of Fourier series is algebraically simpler and more symmetric, when compared to Trigonometric Fourier series. Therefore, it is often used in physics and other sciences.
- ❖ The complex form of Fourier series is useful in problems on electric circuits having impressed periodic voltage.

Complex form of the Fourier series

Definition

The complex exponential form of the Fourier series of a function $f(x)$ defined in the interval $(-\pi, \pi)$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}$$

$$\text{where } c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx.$$

We know that, for a periodic function $f(x)$ of period 2π , defined over $(-\pi, \pi)$, the Fourier series expansion is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \dots \dots \dots (1)$$

We know that $\cos nx = \frac{e^{inx} + e^{-inx}}{2}$ and

Complex form of the Fourier series

$$\sin nx = \frac{e^{inx} - e^{-inx}}{2i} = \frac{-ie^{inx} + ie^{-inx}}{2}$$

Therefore (1) becomes

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \frac{e^{inx} + e^{-inx}}{2} + b_n \frac{-ie^{inx} + ie^{-inx}}{2} \right) \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{a_n - ib_n}{2} \right) e^{inx} + \sum_{n=1}^{\infty} \left(\frac{a_n + ib_n}{2} \right) e^{-inx} \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{a_n - ib_n}{2} \right) e^{inx} + \sum_{n=-\infty}^{-1} \left(\frac{a_{-n} + ib_{-n}}{2} \right) e^{inx} \end{aligned}$$

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Complex form of the Fourier series



$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \text{ where}$$

$$c_n = \begin{cases} \frac{a_0}{2}, & n = 0 \\ \frac{a_n - ib_n}{2}, & n = 1, 2, 3, \dots \\ \frac{a_{-n} + ib_{-n}}{2}, & n = -1, -2, -3, \dots \end{cases}$$

Now for n positive

$$c_n = \frac{a_n - ib_n}{2}$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} (\cos nx - i \sin nx) f(x) dx$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \dots\dots\dots(1)$$

Now for n negative

$$c_n = \frac{a_{-n} + ib_{-n}}{2}$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} (\cos(-nx) + i \sin(-nx)) f(x) dx$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \dots\dots\dots(2)$$

and for $n = 0$

$$c_n = \frac{a_0}{2}$$
$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-0ix} f(x) dx \quad \dots\dots\dots(3)$$

From (1), (2) and (3), we have

$$\therefore c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

Complex form of the Fourier series

Result:

The complex exponential form of the Fourier series of a function $f(x)$ defined on $(-l, l)$ having period $2l$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi x}{l}}$$

where $c_n = \frac{1}{2l} \int_{-l}^l f(x) e^{\frac{-in\pi x}{l}} dx.$

Complex form of the Fourier series - Problems

1. Find the complex exponential form of the Fourier Series of

$$f(x) = \cos ax, \quad -\pi < x < \pi.$$

Solution:

The complex exponential form of the Fourier series is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \dots\dots\dots(1)$$

where

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} \cos ax \, dx$$

Complex form of the Fourier series - Problems

$$c_n = \frac{1}{2\pi} \left[\frac{e^{-inx}}{a^2 - n^2} \{-in \cos ax + a \sin ax\} \right]_{-\pi}^{\pi}$$
$$= \frac{-in \cos a\pi}{2\pi(a^2 - n^2)} [e^{-in\pi} - e^{in\pi}] + \frac{a \sin a\pi}{2\pi(a^2 - n^2)} [e^{-in\pi} + e^{in\pi}]$$

$$c_n = \frac{a \sin a\pi}{2\pi(a^2 - n^2)} 2(-1)^n$$
$$\therefore c_n = \frac{a \sin a\pi}{\pi(a^2 - n^2)} (-1)^n$$

$f(x) = \cos ax = \sum_{-\infty}^{\infty} \frac{a \sin a\pi}{\pi(a^2 - n^2)} (-1)^n e^{inx}$ which is the required Fourier series.



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ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 10

Sub Topic : Problems on Complex Form of the Fourier Series

Department of Science and Humanities

Complex form of the Fourier series - Problems

1. Find the complex exponential form of the Fourier Series of
 $f(x) = e^{-x}$, $-1 < x < 1$

Solution:

The complex exponential form of the Fourier series is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi x}{l}}$$

where

$$c_n = \frac{1}{2l} \int_{-l}^l f(x) e^{\frac{-in\pi x}{l}} dx.$$

Here $l = 1$

Therefore,

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x}$$

Complex form of the Fourier series - Problems

$$c_n = \frac{1}{2} \int_{-1}^1 e^{-x} e^{\frac{-in\pi x}{1}} dx$$

$$c_n = \frac{1}{2} \int_{-1}^1 e^{-(1+in\pi)x} dx$$

$$= \frac{1}{2} \left[\left(\frac{e^{-(1+in\pi)x}}{-(1+in\pi)} \right)_{-1}^1 \right]$$

$$= \frac{1}{2} \left[\frac{e^{-(1+in\pi)x}}{-(1+in\pi)} - \frac{e^{(1+in\pi)x}}{-(1+in\pi)} \right]$$

$$c_n = \frac{1}{2(1+in\pi)} [e e^{in\pi} - e^{-1} e^{in\pi}]$$

Complex form of the Fourier series - Problems

$$\begin{aligned}c_n &= \frac{1}{2(1 + in\pi)} [e(\cos n\pi + i\sin n\pi) - e^{-1}(\cos n\pi - i\sin n\pi)] \\&= \frac{1}{2(1 + in\pi)} (e - e^{-1})(-1)^n \\&= \frac{(1 - in\pi)}{(1 + n^2\pi^2)2} (e - e^{-1})(-1)^n \\c_n &= \frac{(-1)^n(1 - in\pi)\sinh 1}{(1 + n^2\pi^2)}\end{aligned}$$

$$f(x) = e^{-x} = \sum_{-\infty}^{\infty} \frac{(-1)^n(1 - in\pi)\sinh 1}{(1 + n^2\pi^2)} e^{in\pi x}$$

This is the required complex exponential Fourier series

Complex form of the Fourier series - Problems

2. Obtain the complex form of the Fourier series of the function

$$f(x) = \begin{cases} 0 & -\pi \leq x < 0 \\ 1 & 0 < x \leq \pi \end{cases} \text{ and express it as real Fourier series.}$$

Solution

The complex exponential form of the Fourier series is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \dots\dots\dots(1)$$

where

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

$$c_n = \frac{1}{2\pi} \left[\int_{-\pi}^0 0 \cdot e^{\frac{-in\pi x}{\pi}} dx + \int_0^{\pi} 1 \cdot e^{\frac{-in\pi x}{\pi}} dx \right]$$

$$= \frac{1}{2\pi} \int_0^{\pi} e^{-inx} dx$$

$$= \frac{1}{2\pi} \left[\frac{e^{-inx}}{-in} \right]_0^{\pi}$$

$$= \frac{1}{2\pi in} \left[-e^{-in\pi} + 1 \right]$$

$$= \frac{1}{2\pi in} \left[1 - \left(\overset{e^{-in\pi}}{\cos n\pi} - i \overset{0}{\sin n\pi} \right) \right]$$

$$C_n = \frac{1}{2\pi in} [1 - e^{-in\pi}] \text{ is valid when } n \neq 0$$

$$\begin{aligned} 1 - e^{-in\pi} &= 0 \\ 1 - (-1)^n &= 2 \end{aligned}$$

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Complex form of the Fourier series - Problems

When $n=0$

$$C_0 = \frac{1}{2\pi} \int_0^{\pi} 1 \cdot dx$$

$$= \frac{1}{2\pi} [x]_0^{\pi}$$

$$= \frac{1}{2\pi} [\pi]$$

$$\boxed{C_0 = \frac{1}{2}}$$

$$C_n = \begin{cases} 0 & \text{when } n \text{ is even} \\ \frac{1}{\pi i n} & \text{when } n \text{ is odd} \end{cases}$$



From ①

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{inx}$$

$$= \sum_{n=-\infty}^{-1} C_n e^{inx} + C_0 + \sum_{n=1}^{\infty} C_n e^{inx}$$

$$f(x) = \sum_{n=-\infty}^{-1} \frac{1}{\pi i n} e^{inx} + \frac{1}{2} + \sum_{n=1}^{\infty} \frac{1}{\pi i n} e^{inx}$$

$$= \frac{1}{2} + \frac{1}{i\pi} \left[\sum_{n=-\infty}^{-1} \frac{e^{inx}}{n} + \sum_{n=1}^{\infty} \frac{e^{inx}}{n} \right]$$

is the required complex exponential Fourier series

ENGINEERING MATHEMATICS -II

Complex form of the Fourier series - Problems

$$\sin nx = \frac{e^{inx} - e^{-inx}}{2i}$$

$$= \frac{1}{2} + \frac{1}{i\pi} \left[\left(e^{ix} + \frac{e^{i3x}}{3} + \frac{e^{i5x}}{5} + \dots \right) + \left(\frac{e^{-ix}}{-1} + \frac{e^{-i3x}}{-3} + \frac{e^{-i5x}}{-5} + \dots \right) \right]$$

$$= \frac{1}{2} + \frac{1}{i\pi} \left[(e^{ix} - e^{-ix}) + \frac{1}{3} (e^{i3x} - e^{-i3x}) + \frac{1}{5} (e^{i5x} - e^{-i5x}) + \dots \right]$$

$$= \frac{1}{2} + \frac{2}{\pi} \left[\frac{e^{ix} - e^{-ix}}{2i} + \frac{1}{3} \left(\frac{e^{i3x} - e^{-i3x}}{2i} \right) + \frac{1}{5} \left(\frac{e^{i5x} - e^{-i5x}}{2i} \right) + \dots \right]$$

$$f(x) = \frac{1}{2} + \frac{2}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right]$$



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FOURIER TRANSFORMS

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INTRODUCTION



A transform is a mapping between two functions.

Fourier Transform is a type of Integral transform. It converts functions in time domain to frequency domain.

It was discovered by **Jean Baptiste Joseph Fourier** while seeking solution of Heat Equation.

Integral Transform:

An **integral transform** maps a function from its original function space into another function space using integration, where some of the properties of the original function might be more easily characterised and manipulated than in the original function space.

The transformed function can be mapped back to the original function space using the **inverse transform**.

Integral Transform:

The integral transform $F(s)$ of a function $f(x)$ defined by

$$F(s) = \int_a^b f(x) K(s, x) dx$$

provided the integral is convergent and

$K(s, x)$ is called the kernel of the transform.

Depending upon (a, b) and $K(s, x)$ we get different transforms.

When $(a, b) = (0, \infty)$ and $K(s, x) = e^{-sx}$ we get Laplace Transforms

When $(a, b) = (-\infty, \infty)$ and $K(s, x) = e^{isx}$ we get Fourier Transforms

The function $f(x)$ is called the **Inverse transform of $F(s)$** .

INTRODUCTION



Conversion to Frequency domain:

The Fourier Series showed us how to rewrite any periodic function into a sum of sinusoids.

The Fourier Transform is the extension of this idea to non-periodic (aperiodic) functions.

The Fourier Transform is a tool that breaks a function (waveform or signal) into an alternate representation, characterized by sines and cosines.

INTRODUCTION



Dirichlet's Conditions:

The necessary conditions for existence of Fourier Transform are:

The function $f(x)$ should be absolutely integrable i.e.,

$$\int_{-\infty}^{\infty} |f(x)| dx < \infty$$

The function must have finite number of maxima and minima.

The function must have finite number of finite discontinuities.

DEFINITION: FOURIER TRANSFORMATION

Definition: The Fourier Transform of $f(t)$ or complex Fourier transform of $f(t)$ is denoted as $F(\omega)$ and defined as

$$\mathcal{F}[f(t)] = F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt$$



DEFINITION: FOURIER INVERSE TRANSFORMATION



The inverse Fourier transform of $f(t)$ is given by,

$$\begin{aligned}\mathcal{F}^{-1}\{\mathcal{F}[f(t)]\} &= \mathcal{F}^{-1}\{F(\omega)\} \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega\end{aligned}$$

SPECIAL CASE OF FOURIER TRANSFORMATION



By the definition of bilateral Laplace transform

$$L[f(t)] = F(s) = \int_{-\infty}^{\infty} e^{-st} f(t) dt$$

where $s = \sigma + i\omega$.

If we set $\sigma = 0$, we obtain

$$F(\omega) = \int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt$$

which is the Fourier transform of $f(t)$.

FOURIER TRANSFORMATION OF SOME STANDARD FUNCTIONS



$$1. f(t) = e^{iat}$$

By frequency shifting property, we have

$$\mathcal{F}[e^{iat} f(t)] = F(\omega - a)$$

$$\text{Also } \mathcal{F}[1] = 2\pi\delta(\omega)$$

$$\therefore \mathcal{F}[1 \cdot e^{iat}] = \mathcal{F}[e^{iat}] = 2\pi\delta(\omega - a)$$

FOURIER TRANSFORMATION OF SOME STANDARD FUNCTIONS



2. Trigonometric functions, $f(t) = \cos at$

$$\mathcal{F}[\cos at] = \mathcal{F}\left[\frac{e^{iat} + e^{-iat}}{2}\right] = \frac{1}{2}\{\mathcal{F}[e^{iat}] + \mathcal{F}[e^{-iat}]\}$$

$$\mathcal{F}[\cos at] = \frac{1}{2}\{2\pi\delta(\omega - a) + 2\pi\delta(\omega + a)\}$$

$$\begin{aligned}\therefore \mathcal{F}[\cos at] &= \pi\delta(\omega - a) + \pi\delta(\omega + a) \\ &= \pi\{\delta(\omega - a) + \delta(\omega + a)\}\end{aligned}$$

FOURIER TRANSFORMATION OF SOME STANDARD FUNCTIONS



$$3. f(t) = \sin at$$

$$\mathcal{F}[\sin at] = \mathcal{F}\left[\frac{e^{iat} - e^{-iat}}{2i}\right] = \frac{1}{2i} \{\mathcal{F}[e^{iat}] - \mathcal{F}[e^{-iat}]\}$$

$$\begin{aligned}\mathcal{F}[\sin at] &= \frac{1}{2i} \{2\pi\delta(\omega - a) - 2\pi\delta(\omega + a)\} \\ &= \frac{1}{i} \{\pi\delta(\omega - a) - \pi\delta(\omega + a)\}\end{aligned}$$

$$\begin{aligned}\therefore \mathcal{F}[\sin at] &= -i\pi\delta(\omega - a) + i\pi\delta(\omega + a) \\ &= i\pi\{\delta(\omega + a) - \delta(\omega - a)\}\end{aligned}$$



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FOURIER TRANSFORMS

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PROBLEMS:



1. Obtain the Fourier transform of $f(t)$ given by

$$f(t) = \begin{cases} 1 - t^2, & |t| \leq 1 \\ 0, & |t| > 1 \end{cases}$$

Solution: Fourier transform of $f(t)$ is given by

$$\mathcal{F}[f(t)] = F[\omega] = \int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt$$

$f(t) = 1 - t^2$ for $-1 \leq t \leq 1$ and 0 otherwise.

PROBLEMS:



$$[f(t)] = F[\omega] = \int_{-1}^1 (1 - t^2) \cdot e^{-i\omega t} dt$$

$$\mathcal{F}[f(t)] = \mathcal{F} \left[(1 - t^2) \frac{e^{-i\omega t}}{-i\omega} - (-2t) \frac{e^{-i\omega t}}{(-i\omega)^2} + (-2) \frac{e^{-i\omega t}}{(-i\omega)^3} \right]_{-1}^1$$

$$\mathcal{F}[f(t)] = \left[-(1 - t^2) \frac{e^{-i\omega t}}{i\omega} - (2t) \frac{e^{-i\omega t}}{(\omega)^2} - (2) \frac{e^{-i\omega t}}{i\omega^3} \right]_{-1}^1$$

PROBLEMS:



$$\mathcal{F}[f(t)] = \left[0 - \frac{2}{\omega^2} (1 \cdot e^{-i\omega} - (-1) \cdot e^{i\omega}) - \frac{2}{i\omega^3} (e^{-i\omega} - e^{i\omega}) \right]$$

$$\mathcal{F}[f(t)] = \left[-\frac{2}{\omega^2} (e^{-i\omega} + e^{i\omega}) + \frac{2}{i\omega^3} (e^{i\omega} - e^{-i\omega}) \right]$$

$$\mathcal{F}[f(t)] = \left[-\frac{4}{\omega^2} \left(\frac{e^{i\omega} + e^{-i\omega}}{2} \right) + \frac{4}{\omega^3} \left(\frac{e^{i\omega} - e^{-i\omega}}{2i} \right) \right]$$

$$\begin{aligned} \mathcal{F}[f(t)] &= +\frac{4}{\omega^3} \sin \omega \quad \left(\because \cos x = \frac{e^{ix} + e^{-ix}}{2}, \sin x \right. \\ &= \left. \frac{e^{ix} - e^{-ix}}{2i} \right) \end{aligned}$$

$$\therefore \mathcal{F}[f(t)] = 4 \left[\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right]$$

PROBLEMS:

2. Obtain the Fourier transform of $f(t)$ given by

$$f(t) = \begin{cases} 1, & -T \leq t \leq T \\ 0, & |t| > T \end{cases}$$



PROBLEMS:



Solution:

$$\mathcal{F}[f(t)] = F[\omega] = \int_{-T}^T 1 \cdot e^{-i\omega t} dt$$

$$\mathcal{F}[f(t)] = \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-T}^T$$

$$\begin{aligned}\mathcal{F}[f(t)] &= \frac{-1}{i\omega} (e^{-i\omega T} - e^{i\omega T}) \\ &= \frac{1}{i\omega} (e^{i\omega T} - e^{-i\omega T}) \\ &= \frac{2}{\omega} \cdot \frac{(e^{i\omega T} - e^{-i\omega T})}{2i}\end{aligned}$$

$$\therefore \mathcal{F}[f(t)] = \frac{2 \sin \omega T}{\omega}$$



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PROBLEMS:

3. Obtain the Fourier transform of $f(t)$ given by

$$f(t) = \begin{cases} 1 - |t|, & \text{if } |t| \leq 1 \\ 0, & \text{if } |t| > 1 \end{cases}$$

Solution: Fourier transform of $f(t)$ is given by

$$\mathcal{F}[f(t)] = F[\omega] = \int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt$$

$$\mathcal{F}[f(t)] = F[\omega] = \int_{-1}^1 (1 - |t|) \cdot e^{-i\omega t} dt$$

PROBLEMS:



$$\mathcal{F}[f(t)] = \left\{ \int_{-1}^0 1 - (-t) \cdot e^{-i\omega t} dt + \int_0^1 (1 - t) \cdot e^{-i\omega t} dt \right\}$$

$$\mathcal{F}[f(t)] = \int_{-1}^0 (1 + t) \cdot e^{-i\omega t} dt + \int_0^1 (1 - t) \cdot e^{-i\omega t} dt$$

$$= \left(\left[(1 + t) \frac{e^{-i\omega t}}{-i\omega} \right]_{-1}^0 - \int_{-1}^0 1 \cdot \frac{e^{-i\omega t}}{-i\omega} dt \right) + \left(\left[(1 - t) \frac{e^{-i\omega t}}{-i\omega} \right]_0^1 - \int_0^1 (-1) \cdot \frac{e^{-i\omega t}}{-i\omega} dt \right)$$

$$= \left(-\frac{1}{i\omega} (1 - 0) - \left[\frac{e^{-i\omega t}}{(-i\omega)^2} \right]_{-1}^0 \right) + \left(-\frac{1}{i\omega} (0 - 1) + \left[\frac{e^{-i\omega t}}{(-i\omega)^2} \right]_0^1 \right)$$

PROBLEMS:



$$\mathcal{F}[f(t)] = \left[-\frac{1}{i\omega} + \frac{1}{\omega^2} (1 - e^{i\omega}) \right] + \left[\frac{1}{i\omega} - \frac{1}{\omega^2} (e^{-i\omega} - 1) \right]$$

$$= \left\{ \frac{1}{\omega^2} - \frac{1}{\omega^2} e^{i\omega} - \frac{1}{\omega^2} e^{-i\omega} + \frac{1}{\omega^2} \right\} = \frac{2}{\omega^2} - \frac{1}{\omega^2} (e^{i\omega} + e^{-i\omega})$$

$$= \frac{2}{\omega^2} - \frac{2}{\omega^2} \cdot \frac{(e^{i\omega} + e^{-i\omega})}{2}$$

$$\mathcal{F}[f(t)] = \frac{2}{\omega^2} - \frac{2 \cos \omega}{\omega^2}$$

$$\therefore \mathcal{F}[f(t)] = \frac{2 - 2 \cos \omega}{\omega^2}$$

PROBLEMS:



4. Obtain the Fourier transform of $f(t) = \begin{cases} t^2, & |t| < a \\ 0, & |t| > a \end{cases}$, where a is positive constant.

Solution: Fourier transform of $f(t)$ is given by

$$\mathcal{F}[f(t)] = F[\omega] = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

Since $f(t) = t^2$ for $|t| < a$ by data we have

$$F[\omega] = \int_{-a}^a t^2 e^{-i\omega t} dt$$

PROBLEMS:

$$\begin{aligned} F[\omega] &= \left[(t^2) \frac{e^{-i\omega t}}{-i\omega} - (2t) \frac{e^{-i\omega t}}{(-i\omega)^2} + (2) \frac{e^{-i\omega t}}{(-i\omega)^3} \right]_{-a}^a \\ &= -\frac{1}{i\omega} (a^2 e^{-i\omega a} - a^2 e^{i\omega a}) + \frac{2}{\omega^2} (a e^{-i\omega a} + a e^{i\omega a}) + \\ &\quad \frac{2}{i\omega^3} (e^{-i\omega a} - e^{i\omega a}) \\ &= \frac{a^2}{i\omega} (e^{i\omega a} - e^{-i\omega a}) + \frac{2a}{\omega^2} (e^{i\omega a} + e^{-i\omega a}) \\ &\quad - \frac{2}{i\omega^3} (e^{i\omega a} - e^{-i\omega a}) \\ &= \frac{a^2}{i\omega} (2i \sin a\omega) + \frac{2a}{\omega^2} (2 \cos a\omega) - \frac{2}{i\omega^3} (2i \sin a\omega) \\ &\left(\because \cos x = \frac{e^{ix} + e^{-ix}}{2}, \sin x = \frac{e^{ix} - e^{-ix}}{2i} \right) \end{aligned}$$

PROBLEMS:



$$F(\omega) = \frac{2 a^2 \sin a \omega}{\omega} + \frac{4 a \cos a \omega}{\omega^2} - \frac{4 \sin a \omega}{\omega^3}$$

$$= \frac{2 a^2 \omega^2 \sin a \omega + 4 a \omega \cos a \omega - 4 \sin a \omega}{\omega^3}$$

$$\therefore F(\omega) = \frac{1}{\omega^3} [2(a^2 \omega^2 - 2) \sin a \omega + 4 a \omega \cos a \omega]$$



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Fourier Sine Transform:



Definition:

The Fourier Sine Transform for the function $f(t)$ is defined by

$$\mathcal{F}_s[f(t)] = F_s(\omega) = \int_0^{\infty} f(t) \sin \omega t dt$$

Definition:

The Inverse Fourier Sine Transform for the function $F_s(\omega)$ is defined by

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_s(\omega) \sin \omega t d\omega$$

Fourier Sine Transforms:



Problem 1: Obtain Fourier sine Transform of $f(t) = \begin{cases} 4t, & 0 < t < 1 \\ 4-t, & 1 < t < 4 \\ 0, & t > 4 \end{cases}$

Solution:

$$\begin{aligned}\mathcal{F}_s[f(t)] &= \int_0^{\infty} f(t) \sin \omega t dt \\ &= \int_0^1 4t \sin \omega t dt + \int_1^4 (4-t) \sin \omega t dt \\ &= 4 \left[t \frac{(-\cos \omega t)}{\omega} - \frac{(-\sin \omega t)}{\omega^2} \right]_0^1 + \left[(4-t) \frac{(-\cos \omega t)}{\omega} - \frac{(-\sin \omega t)}{\omega^2} (-1) \right]_1^4 \\ &= 4 \left[\frac{(-\cos \omega)}{\omega} + \frac{(\sin \omega)}{\omega^2} \right] + \left[\frac{-(-3\cos \omega)}{\omega} - \frac{(\sin 4\omega - \sin \omega)}{\omega^2} \right] \\ &= \frac{-(\cos \omega)}{\omega} + \frac{(-\sin 4\omega + 5\sin \omega)}{\omega^2}\end{aligned}$$

Fourier Sine Transforms:



Problem 2: Obtain Fourier sine Transform of $f(t) = \frac{e^{-at}}{t}, a > 0$.

Solution:
$$\mathcal{F}_s[f(t)] = \int_0^{\infty} f(t) \sin \omega t dt = \int_0^{\infty} \frac{e^{-at}}{t} \sin \omega t dt = F_s(\omega) \text{ --- (1)}$$

Differentiating wrt ω and applying Leibnitz rule of integration under integral sign, we get,

$$\frac{dF_s(\omega)}{d\omega} = \int_0^{\infty} \frac{e^{-at}}{t} t \cos \omega t dt = \int_0^{\infty} e^{-at} \cos \omega t dt$$

Using the standard result, $\int e^{at} \cos btdt = \frac{e^{at}}{a^2 + b^2} (a \cos bt + b \sin bt)$

$$\frac{dF_s(\omega)}{d\omega} = \left(\frac{e^{-at}}{a^2 + \omega^2} (-a \cos \omega t + \omega \sin \omega t) \right) \Bigg|_0^{\infty}$$

Fourier Sine Transforms:



$$\Rightarrow \frac{dF_s(\omega)}{d\omega} = \left(0 - \frac{1}{a^2 + \omega^2} (-a) \right) = \frac{a}{a^2 + \omega^2}$$

$$\Rightarrow F_s(\omega) = \int \frac{a}{a^2 + \omega^2} d\omega + c = \tan^{-1} \left(\frac{\omega}{a} \right) + c$$

$$\Rightarrow F_s(0) = c$$

By (1), $F_s(0) = 0$

Therefore, $F_s(\omega) = \tan^{-1} \left(\frac{\omega}{a} \right)$

$$\left(F_s(\omega) = \int_0^{\infty} \frac{e^{-at}}{t} \sin \omega t dt - - - (1) \right)$$

Problem 3: Find Inverse Fourier Sine Transform of $\frac{\omega}{1+\omega^2}$

Solution: $f(t) = \frac{2}{\pi} \int_0^{\infty} F_s(\omega) \sin \omega t d\omega = \frac{2}{\pi} \int_0^{\infty} \left(\frac{\omega}{1+\omega^2} \right) \sin \omega t d\omega \quad \text{--- (1)}$

$$= \frac{2}{\pi} \int_0^{\infty} \left(\frac{\omega^2}{1+\omega^2} \right) \frac{\sin \omega t}{\omega} d\omega$$

$$= \frac{2}{\pi} \int_0^{\infty} \left(\frac{1+\omega^2-1}{1+\omega^2} \right) \frac{\sin \omega t}{\omega} d\omega$$

$$= \frac{2}{\pi} \left[\int_0^{\infty} \frac{\sin \omega t}{\omega} d\omega - \int_0^{\infty} \left(\frac{1}{1+\omega^2} \right) \frac{\sin \omega t}{\omega} d\omega \right]$$

Fourier Sine Transforms :



$$f(t) = \frac{2}{\pi} \left[\frac{\pi}{2} - \int_0^{\infty} \frac{1}{\omega(1+\omega^2)} \sin \omega t d\omega \right]$$

$$f'(t) = -\frac{2}{\pi} \int_0^{\infty} \frac{\omega}{\omega(1+\omega^2)} \cos \omega t d\omega = -\frac{2}{\pi} \int_0^{\infty} \frac{1}{(1+\omega^2)} \cos \omega t d\omega$$

$$f''(t) = +\frac{2}{\pi} \int_0^{\infty} \frac{\omega}{(1+\omega^2)} \sin \omega t d\omega = \frac{2}{\pi} \frac{\pi}{2} f(t)$$

$$\Rightarrow f''(t) = f(t)$$

$$\Rightarrow (D^2 - 1) f(t) = 0$$

$$\Rightarrow f(t) = C_1 e^t + C_2 e^{-t} \text{ --- (2)}$$

Fourier Sine Transforms :



$$f'(t) = C_1 e^t - C_2 e^{-t} \text{ --- (3)}$$

Put $t=0$ in (2) and (3) we get,

$$f(0) = C_1 + C_2 \text{ and } f'(0) = C_1 - C_2$$

i.e, $C_1 + C_2 = 1 \text{ and } C_1 - C_2 = -1$

On solving , we get, $C_1 = 0 \text{ and } C_2 = 1$

Thus, $f(t) = e^{-t}$



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Problem 4: Solve the integral equation $\int_0^{\infty} f(x) \sin(xt) dx = \begin{cases} 1, & 0 < t < 1 \\ 2, & 1 < t < 2 \\ 0, & \text{otherwise} \end{cases}$

Solution: We know that $F_s(\omega) = \int_0^{\infty} f(t) \sin \omega t dt = \mathcal{F}_s[f(x)]$

$$\text{Given that, } \int_0^{\infty} f(x) \sin xtdt = \begin{cases} 1, & 0 < t < 1 \\ 2, & 1 < t < 2 \\ 0, & \text{otherwise} \end{cases} = \mathcal{F}_s[f(x)]$$

$$\text{i.e., } F_s(\omega) = \begin{cases} 1, & 0 < \omega < 1 \\ 2, & 1 < \omega < 2 \\ 0, & \text{otherwise} \end{cases} = \mathcal{F}[f(t)]. \text{ Therefore, in order to find } f(t) \text{ we need to find inverse transform of } F_s(\omega)$$

$$\text{Thus, } f(t) = \frac{2}{\pi} \int_0^{\infty} F_s(\omega) \sin \omega t d\omega$$

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Fourier Sine Transforms :



$$= \frac{2}{\pi} \left[\int_0^1 \sin \omega t d\omega + \int_1^2 2 \sin \omega t d\omega \right]$$

$$= \frac{2}{\pi} \left[\left\{ \frac{-\cos \omega t}{t} \right\}_0^1 + \left\{ \frac{-2 \cos \omega t}{t} \right\}_1^2 \right]$$

$$= \frac{2}{\pi} \left[\left\{ \frac{-(\cos t - 1)}{t} \right\} + \left\{ \frac{-2(\cos 2t - \cos t)}{t} \right\} \right]$$

$$f(t) = \frac{2}{\pi} \left[\left\{ \frac{1 - 2 \cos 2t + \cos t}{t} \right\} \right]$$

Fourier Sine Transforms:



Problem 5: Find Fourier Sine Transform of e^{-at} ($a > 0$) and hence show that $\int_0^{\infty} \frac{x \sin kx}{x^2 + a^2} dx = \frac{\pi}{2} e^{-ak}$

Solution: $\mathcal{F}_s[f(t)] = \int_0^{\infty} f(t) \sin \omega t dt \Rightarrow \mathcal{F}_s[e^{-at}] = \int_0^{\infty} e^{-at} \sin \omega t dt$

$$= \left[\frac{e^{-at}}{a^2 + \omega^2} \{-a \sin \omega t - \omega \cos \omega t\} \right]_0^{\infty}$$

$$= \frac{\omega}{a^2 + \omega^2} = F_s(\omega)$$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_s(\omega) \sin \omega t d\omega$$

$$= \frac{2}{\pi} \int_0^{\infty} \frac{\omega}{a^2 + \omega^2} \sin \omega t d\omega$$

Fourier Sine Transforms:



$$\Rightarrow e^{-at} = \frac{2}{\pi} \int_0^{\infty} \frac{\omega \sin \omega t}{a^2 + \omega^2} d\omega$$

$$t = k \Rightarrow e^{-ak} \frac{\pi}{2} = \int_0^{\infty} \frac{\omega \sin \omega k}{a^2 + \omega^2} d\omega;$$

$$\omega \rightarrow x \Rightarrow \boxed{\int_0^{\infty} \frac{x \sin kx}{a^2 + x^2} dx = \frac{\pi}{2} e^{-ak}}$$

Fourier Sine Transforms:



Fourier sine transform of derivatives:

Let $f(t)$ and $f'(t)$ be continuous on $[0, \infty)$. Let $f(t) \rightarrow 0$ and $f'(t) \rightarrow 0$ as $t \rightarrow \infty$ and $f''(t)$ be piecewise continuous on every subinterval $[0, l]$. Then,

$$\mathcal{F}_s[f''(t)] = -\omega^2 F_s(f(t)) + \omega f(0)$$

Proof: By definition we have,

$$\begin{aligned}\mathcal{F}_s[f''(t)] &= \int_0^{\infty} f''(t) \sin \omega t dt \\ &= \left[\sin \omega t f'(t) \right]_0^{\infty} - \int_0^{\infty} \omega \cos \omega t f'(t) dt\end{aligned}$$

Fourier Sine Transforms:



$$= 0 - \omega \int_0^{\infty} \cos \omega t f'(t) dt = -\omega \left\{ [\cos \omega t f(t)]_0^{\infty} + \omega \int_0^{\infty} \sin \omega t f(t) dt \right\}$$

$$= -\omega \left\{ 0 - f(0) + \omega \int_0^{\infty} \sin \omega t f(t) dt \right\}$$

$$\mathcal{F}_s[f''(t)] = -\omega^2 F_s(f(t)) + \omega f(0)$$



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Definition:

The Fourier Cosine Transform for the function $f(t)$ is defined by

$$\mathcal{F}_c[f(t)] = F_c(\omega) = \int_0^{\infty} f(t) \cos \omega t dt$$

Definition:

The Inverse Fourier Cosine Transform for the function $F_c(\omega)$ is defined by

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_c(\omega) \cos \omega t d\omega$$

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Fourier Cosine Transforms:



Problem 1: Find Fourier Cosine Transform of e^{-at} ($a > 0$) and hence evaluate

$$(i) \int_0^{\infty} \frac{\cos \lambda x}{x^2 + a^2} dx \quad ii) \mathcal{F}_c(te^{-at})]$$

Solution: We have,

$$\mathcal{F}_c[f(t)] = \int_0^{\infty} f(t) \cos \omega t dt = F_c(\omega)$$

$$\mathcal{F}_c[e^{-at}] = \int_0^{\infty} e^{-at} \cos \omega t dt$$

$$= \left[\frac{e^{-at}}{a^2 + \omega^2} \{-a \cos \omega t + \omega \sin \omega t\} \right]_0^{\infty}$$

$$F_c(\omega) = \frac{a}{a^2 + \omega^2}$$

$$(i) f(t) = \frac{2}{\pi} \int_0^{\infty} F_c(\omega) \cos \omega t d\omega$$

$$\Rightarrow e^{-at} = \frac{2}{\pi} \int_0^{\infty} \frac{a}{a^2 + \omega^2} \cos \omega t d\omega$$

$$t = \lambda \Rightarrow e^{-a\lambda} \frac{\pi}{2a} = \int_0^{\infty} \frac{\cos \omega \lambda}{a^2 + \omega^2} d\omega;$$

$$\omega \rightarrow x \Rightarrow \int_0^{\infty} \frac{\cos x \lambda}{a^2 + x^2} dx = \frac{\pi}{2a} e^{-a\lambda}$$

$$ii) \mathcal{F}_c[e^{-at}] = \frac{a}{a^2 + \omega^2} = F_c(\omega)$$

$$\frac{d}{da} \{ \mathcal{F}_c[e^{-at}] \} = \frac{d}{da} \left(\frac{a}{a^2 + \omega^2} \right) \Rightarrow \frac{d}{da} \left(\int_0^{\infty} e^{-at} \cos \omega t dt \right) = \frac{d}{da} \left(\frac{a}{a^2 + \omega^2} \right)$$

$$\Rightarrow \left(\int_0^{\infty} \frac{\partial}{\partial a} \{ e^{-at} \cos \omega t dt \} \right) = \frac{a^2 + \omega^2 - a(2a)}{(a^2 + \omega^2)^2}$$

$$\Rightarrow \int_0^{\infty} -te^{-at} \cos \omega t dt = \frac{\omega^2 - a^2}{(a^2 + \omega^2)^2}$$

$$\Rightarrow F_c[te^{-at}] = \frac{a^2 - \omega^2}{(a^2 + \omega^2)^2}$$

Problem 2: Find Fourier Cosine Transform of t^{n-1} ($n \geq 2$) and hence find $\mathcal{F}_c \left[\frac{1}{\sqrt{t}} \right]$

Solution: $\mathcal{F}_c[f(t)] = \int_0^{\infty} f(t) \cos \omega t dt$

i. e., $\mathcal{F}_c[t^{n-1}] = \int_0^{\infty} t^{n-1} \cos \omega t dt$

$$= \operatorname{Re} \int_0^{\infty} t^{n-1} e^{-i\omega t} dt = \operatorname{Re} \frac{\Gamma(n)}{(i\omega)^n}$$

$$= \operatorname{Re} \frac{\Gamma(n)}{i^n \omega^n} = \operatorname{Re} \frac{\Gamma(n)}{\omega^n} (-i)^n$$

$$= \operatorname{Re} \frac{\Gamma(n)}{\omega^n} \left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right)^n$$

$$\left(\because \int_0^{\infty} x^{n-1} e^{-ax} dx = \frac{\Gamma(n)}{(a)^n} \right)$$

$$\mathcal{F}_c[t^{n-1}] = \frac{\Gamma(n)}{\omega^n} \cos \frac{n\pi}{2} = \frac{n!}{\omega^n} \cos \frac{n\pi}{2}$$

$$n = \frac{1}{2} \Rightarrow \mathcal{F}_c \left[\frac{1}{\sqrt{t}} \right] = \frac{\Gamma\left(\frac{1}{2}\right)}{\omega^{\frac{1}{2}}} \cos \frac{\pi}{4}$$

$$= \frac{\sqrt{\pi}}{\sqrt{\omega}} \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{\pi}}{\sqrt{2}} \frac{1}{\sqrt{\omega}}$$

Problem 3: Solve the integral equation $\int_0^{\infty} f(\theta) \cos \alpha \theta d\theta = \begin{cases} 1-\alpha, & 0 \leq \alpha \leq 1 \\ 0, & \text{otherwise} \end{cases}$

and hence evaluate $\int_0^{\infty} \frac{\sin^2 t}{t^2} dt$

Solution:

$$F_c(\alpha) = \begin{cases} 1-\alpha, & 0 \leq \alpha \leq 1 \\ 0, & \text{otherwise} \end{cases} \quad f(\theta) = ?$$

$$f(\theta) = \frac{2}{\pi} \int_0^{\infty} F_c(\alpha) \cos \alpha \theta d\alpha = \frac{2}{\pi} \int_0^1 (1-\alpha) \cos \alpha \theta d\alpha$$

$$= \frac{2}{\pi} \left[(1-\alpha) \frac{\sin \alpha \theta}{\theta} - \frac{(-\cos \alpha \theta)}{\theta^2} (-1) \right]_0^1$$

$$f(\theta) = \frac{2}{\pi} \frac{-(\cos \theta - 1)}{\theta^2} = \frac{2}{\pi} \frac{(1 - \cos \theta)}{\theta^2}$$

$$\int_0^{\infty} f(\theta) \cos \alpha \theta d\theta = \frac{2}{\pi} \int_0^{\infty} \frac{(1-\cos \theta)}{\theta^2} \cos \alpha \theta d\theta = F_c(\alpha);$$

$$\alpha = 0 \Rightarrow F_c(\alpha=0) = 1$$

$$\Rightarrow \frac{2}{\pi} \int_0^{\infty} \frac{(1-\cos \theta)}{\theta^2} d\theta = 1 \Rightarrow \int_0^{\infty} \frac{(1-\cos \theta)}{\theta^2} d\theta = \frac{\pi}{2}$$

$$\text{Put } \theta = 2t \Rightarrow d\theta = 2dt$$

$$\Rightarrow \int_0^{\infty} \frac{(1-\cos 2t)}{4t^2} 2dt = \frac{\pi}{2}$$

$$\therefore \int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$$



THANK YOU

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ENGINEERING MATHEMATICS-II UE22MA141B

INVERSE FOURIER COSINE TRANSFORM

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Recall

The Fourier cosine transform of a real function is the real part of the full complex Fourier transform

DEFINITION

If $f(t)$ be defined in $0 \leq t < \infty$ and $f(t)$ is extended as even function in $(-\infty, \infty)$ satisfying the condition on Fourier integral theorem .

The Fourier cosine transform for the function $f(t)$ is defined by

$$F_c(\omega) = \int_0^{\infty} f(t) \cos \omega t \, dt$$

where ' ω ' is a parameter

The Fourier sine and cosine transforms are forms of the Fourier transform that do not use complex numbers or require negative frequency. They are the forms originally used by Joseph Fourier and are still preferred in some applications, such as signal processing or statistics



INVERSE FOURIER COSINE TRANSFORM

If $F_C(\omega)$ is the Fourier cosine transform of $f(t)$ which satisfies Dirichlet's condition in every finite interval $(0, l)$

Then

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_C(\omega) \cos \omega t \, d\omega$$

$f(t)$ is called inverse cosine Fourier transform of $F_C(\omega)$

EXAMPLE 1

Find the inverse Fourier cosine transform of $e^{-\omega}$

$$\mathcal{F}_C(\omega) = e^{-\omega}$$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_C(\omega) \cos \omega t d\omega$$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} e^{-\omega} \cos \omega t d\omega$$

$$\text{w.k.t } \int_0^{\infty} e^{-a\omega} \cos \omega t d\omega = \frac{a}{a^2 + t^2} \text{ and } \int_0^{\infty} e^{-a\omega} \sin \omega t d\omega = \frac{t}{a^2 + t^2}$$

$$f(t) = \frac{2}{\pi} \left(\frac{1}{1^2 + t^2} \right)$$

EXAMPLE 2

Find $f(t)$ if its Fourier cosine transform is $\mathcal{F}_c(\omega) = \begin{cases} k - \frac{\omega}{2} & \text{for } \omega < 2k \\ 0 & \text{for } \omega > 2k \end{cases}$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_c(\omega) \cos \omega t d\omega$$

$$f(t) = \frac{2}{\pi} \int_0^{2k} \left(k - \frac{\omega}{2}\right) \cos \omega t d\omega + \int_{2k}^{\infty} (0) \cos \omega t d\omega$$

Integration by parts

$$f(t) = \frac{2}{\pi} \left[\left(k - \frac{\omega}{2} \right) \frac{\sin \omega t}{t} \right]_{\omega=0}^{\omega=2k} - \left[\left(-\frac{1}{2} \right) \frac{-\cos \omega t}{t^2} \right]_{\omega=0}^{\omega=2k}$$

$$f(t) = \frac{2}{\pi} \left[\left[\left(k - \frac{2k}{2} \right) \frac{\sin 2kt}{t} - \left(k - \frac{0}{2} \right) \frac{\sin 0}{t} \right] - \frac{1}{2} \left[\frac{\cos 2kt}{t^2} - \frac{\cos 0}{t^2} \right] \right]$$

$$f(t) = \frac{2}{\pi} \left[\frac{-1}{2} \left[\frac{\cos 2kt}{t^2} - \frac{1}{t^2} \right] \right]$$

$$f(t) = \frac{2}{\pi} \left[\frac{\sin^2 kt}{t^2} \right]$$



Thank you

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ENGINEERING MATHEMATICS-II UE22MA141B

Properties of Fourier Transform -1

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Linearity property:

If a and b are any 2 constants and $f(t)$ and $g(t)$ are any 2 functions whose Fourier transforms exist, then

$$\mathcal{F}[a f(t) + b g(t)] = a \mathcal{F}[f(t)] + b \mathcal{F}[g(t)].$$

Proof: By definition of Fourier transform

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$\mathcal{F}[a f(t) + b g(t)] = \int_{-\infty}^{\infty} \{a f(t) + b g(t)\} e^{-i\omega t} dt$$

$$\mathcal{F}[a f(t) + b g(t)] = a \left[\int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \right] + b \left[\int_{-\infty}^{\infty} g(t) e^{-i\omega t} dt \right]$$

$$\therefore F[a f(t) + b g(t)] = a \mathcal{F}[f(t)] + b \mathcal{F}[g(t)] = a F(\omega) + b G(\omega)$$

-
- Addition of two functions corresponding to the addition of the two frequency spectrum is called the linearity. If we multiply a function by a constant, the Fourier transform of the resultant function is multiplied by the same constant. The Fourier transform of sum of two or more functions is the sum of the Fourier transforms of the functions.

- **Case I.** If $h(x) \rightarrow H(\omega)$ then $ah(x) \rightarrow aH(\omega)$

- **Case II.** If $h(x) \rightarrow H(\omega)$ and $g(x) \rightarrow G(\omega)$ then

$$h(x)+g(x) \rightarrow H(\omega) +G(\omega)$$

EXAMPLE

Find Fourier transform of $y(t) = e^{-a|t|}$.

Solution

given input can be written in terms of unit step function

$$y(t) = e^{-at}u(t) + e^{at}u(-t)$$

$\mathcal{F}[y(t)] = \mathcal{F}[e^{-at}u(t) + e^{at}u(-t)]$ by Linearity property

$$= \mathcal{F}[e^{-at}u(t)] + \mathcal{F}[e^{at}u(-t)] = \frac{1}{a+i\omega} + \frac{1}{a-i\omega} = \frac{2a}{a^2+\omega^2}$$

Shifting property:

1. If a is a real constant then

$$\mathcal{F}[f(t - a)] = e^{-i\omega a} F(\omega)$$

Proof:

$$\mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$\mathcal{F}[f(t - a)] = \int_{-\infty}^{\infty} f(t - a) e^{-i\omega t} dt$$

Let $s = t - a \Rightarrow t = s + a$

$$ds = dt$$

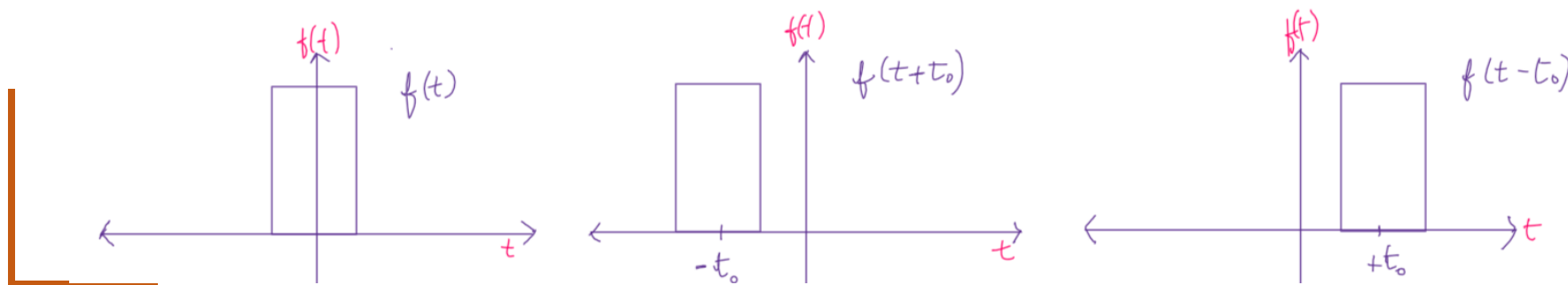
If $t \rightarrow -\infty, s \rightarrow -\infty$ and if $t \rightarrow \infty, s \rightarrow \infty$

$$\therefore \mathcal{F}[f(t - a)] = \int_{-\infty}^{\infty} f(s) e^{-i\omega(a+s)} ds$$

$$\begin{aligned} &= e^{-i\omega a} \int_{-\infty}^{\infty} f(s) e^{-i\omega s} ds \\ &= e^{-i\omega a} F(\omega) \end{aligned}$$

When the signal is shifted in time domain is said to be delayed or advanced based on whether the signal is shifted to right or left.

For example



Example

find the inverse Fourier transform of $\frac{e^{3i\omega}}{2-i\omega}$

$$\text{wkt } \mathcal{F}[e^{-at}u(t)] = \frac{1}{a+i\omega}$$

$$\text{and } \mathcal{F}[e^{at}u(-t)] = \frac{1}{a-i\omega}$$

$$\mathcal{F}^{-1}\left[\frac{1}{2-i\omega}\right] = e^{2t}u(-t) = f(t)$$

$$\mathcal{F}^{-1}\left[e^{-i\omega a}F(\omega)\right] = f(t - a)$$

$$\mathcal{F}^{-1}\left[\frac{e^{3i\omega}}{2 - i\omega}\right] = f(t + 3) = e^{2(t+3)}u(-(t + 3))$$

Frequency shifting property:

If $\mathcal{F}[f(t)] = F(\omega)$ then $\mathcal{F}[e^{iat}f(t)] = F(\omega - a)$

Proof:

$$\mathcal{F}[f(t)e^{iat}] = \int_{-\infty}^{\infty} f(t) \cdot e^{iat} \cdot e^{-i\omega t} dt$$

$$\mathcal{F}[f(t)e^{iat}] = \int_{-\infty}^{\infty} f(t) \cdot e^{iat-i\omega t} dt = \int_{-\infty}^{\infty} f(t) \cdot e^{-i(\omega-a)t} dt$$

$$\therefore \mathcal{F}[f(t)e^{iat}] = F(\omega - a)$$

Frequency is shifted according to the co-ordinates. There is a duality between the time and frequency domains and frequency shift affects the time shift.

If $f(t) \rightarrow F(\omega)$ then $f(t) \exp[i\omega' t] \rightarrow F(\omega - \omega')$ Shifting the frequency by ω' in frequency domain is equivalent to multiplying the time domain signal by $e^{i\omega' t}$



Modulation Theorem

If $\mathcal{F}[f(t)] = F(\omega)$ and ω_0 is any real number, then

$$\mathcal{F}[f(t) \cos(\omega_0 t)] = \frac{1}{2} [F(\omega + \omega_0) + F(\omega - \omega_0)]$$

and

$$\mathcal{F}[f(t) \sin(\omega_0 t)] = \frac{i}{2} [F(\omega + \omega_0) - F(\omega - \omega_0)]$$

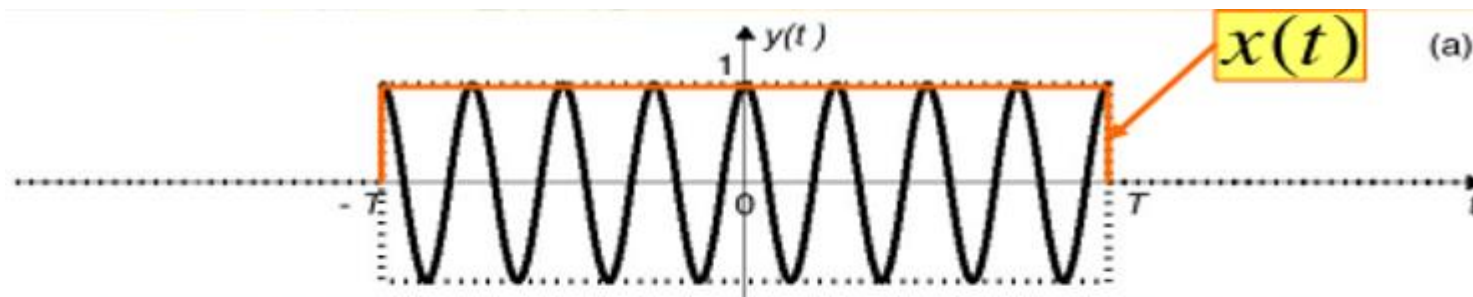
Proof w .k. t $f(t) \cos(\omega_0 t) = \frac{f(t)}{2} [e^{i\omega_0 t} + e^{-i\omega_0 t}]$

$$\mathcal{F} \left[\frac{f(t)}{2} [e^{i\omega_0 t} + e^{-i\omega_0 t}] \right] = \frac{\mathcal{F}[f(t)[e^{i\omega_0 t}]]}{2} + \frac{\mathcal{F}[f(t)[e^{-i\omega_0 t}]]}{2} = \frac{F(\omega - \omega_0)}{2} + \frac{F(\omega + \omega_0)}{2}$$

If $\mathcal{F}[f(t)] = F(\omega)$ then $\mathcal{F}[e^{iat} f(t)] = F(\omega - a)$ (frequency shift property)

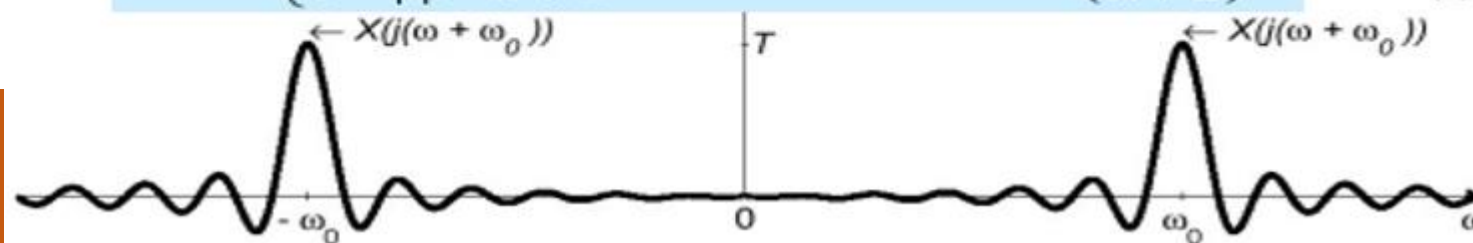
$$y(t) = x(t) \cos(\omega_0 t) \Leftrightarrow$$

$$Y(\omega) = \frac{1}{2} X(\omega - \omega_0) + \frac{1}{2} X(\omega + \omega_0)$$



$$x(t) = \begin{cases} 1 & |t| < T/2 \\ 0 & |t| > T/2 \end{cases} \Leftrightarrow X(\omega) = \frac{\sin(\omega T / 2)}{(\omega / 2)}$$

(b)





Thank you

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ENGINEERING MATHEMATICS-II UE22MA141B

Properties of Fourier Transform -2

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Fourier Transform of Derivatives

Let $f(t)$ be continuous and $f^{(k)}(t)$, $k = 1, 2, \dots, n$ be piecewise continuous on every interval $[-l, l]$ and

$\int_{-\infty}^{\infty} |f^{(k-1)}(t)| dt$, $k = 1, 2, \dots, n$ converges. Let $f^{(k)}(t) \rightarrow 0$

as $t \rightarrow \infty$ for $k = 0, 1, 2, \dots, n - 1$. If $\mathcal{F}[f(t)] = F(\omega)$, then

$$\mathcal{F}[f^{(n)}(t)] = (i\omega)^n F(\omega),$$

Where $f(t)$ and all its derivatives vanish at infinity.

For Example,

$$\mathcal{F}[f'(t)] = \int_{-\infty}^{\infty} f'(t)e^{-i\omega t} dt$$

$$= [f(t)e^{-i\omega t}]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} i\omega f(t)e^{-i\omega t} dt = i\omega F(\omega).$$

$$\mathcal{F}[f''(t)] = -\omega^2 F(\omega).$$

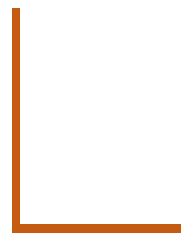
Example

Solve differential equation using Fourier transform

$$f' - 4f = u(t)e^{-4t}$$

$$i\omega F(\omega) - 4F(\omega) = \frac{1}{4 + i\omega}$$

$$(i\omega - 4)F(\omega) = \frac{1}{4 + i\omega}$$



$$F(\omega) = \frac{1}{(4 + i\omega)(i\omega - 4)}$$

$$F(\omega) = \frac{-1}{(4 + i\omega)(4 - i\omega)}$$

$$F(\omega) = -\frac{1}{4^2 + \omega^2} \quad \left\{ \frac{2a}{a^2 + \omega^2} = \mathcal{F}[e^{-a|t|}] \right\}$$

$$f(t) = -\frac{1}{2 \cdot 4} e^{-4|t|} = -\frac{1}{8} e^{-4|t|}$$

Symmetry property of Fourier transforms

Let $\mathcal{F}[f(t)] = F(\omega)$. Then

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

$$f(t) = \mathcal{F}^{-1}[\omega] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

$$2\pi f(t) = \int_{-\infty}^{\infty} F(s) e^{ist} ds$$

$$\begin{aligned} 2\pi f(-\omega) &= \int_{-\infty}^{\infty} F(s) e^{-is\omega} ds \quad \text{change } s \rightarrow t \\ &= \int_{-\infty}^{\infty} F(t) e^{-i\omega t} dt = \mathcal{F}[F(t)] \end{aligned}$$

Example

Find Fourier transform of $\frac{1}{2\pi} \text{sinc} t$

Solution

We know that $F(\omega) = \frac{1}{\sqrt{2\pi}} \text{sinc} \omega$ for

$$f(t) = \begin{cases} 0 & t < -1 \\ 1 & -1 < t < 1 \\ 0 & t > 1 \end{cases} \quad \text{typical box wave}$$

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

$$\mathcal{F}\left(\frac{1}{\sqrt{2\pi}} \text{sinc} t\right) = \mathcal{F}(F(t)) =$$

$$2\pi f(-\omega) = \begin{cases} 0 & -\omega < -1 \\ 2\pi & -1 < -\omega < 1 \\ 0 & -\omega > 1 \end{cases}$$

Differential with respect to frequency (ω)

If $\mathcal{F}(f(t)) = F(\omega)$ then

$$\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$$

Differentiating with respect to ω

$$F'(\omega) = \frac{d}{d\omega} \left(\int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt \right)$$


$$F'(\omega) = \int_{-\infty}^{\infty} \frac{d}{d\omega} (f(t) \cdot e^{-i\omega t}) dt$$

$$F'(\omega) = \int_{-\infty}^{\infty} -it \cdot f(t) \cdot e^{-i\omega t} dt$$

$$F'(\omega) = -i[\mathcal{F}(tf(t))]$$

Similarly

$$F''(\omega) = (-i)^2[\mathcal{F}(t^2f(t))]$$

$$\text{Therefore } \mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$$

$$(-i)^n t^n f(t) = \mathcal{F}^{-1}(F^{(n)}(\omega))$$

Example

Obtain Fourier transform for $[te^{-4t}u(t)]$.

solution : w.K.t $\mathcal{F}[e^{-at}u(t)] = \frac{1}{a+i\omega}$

And from previous property $\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$

$$\mathcal{F}[e^{-4t}u(t)] = \frac{1}{4 + i\omega}$$

$$\mathcal{F}[te^{-4t}u(t)] = i \cdot \frac{d}{d\omega} \left[\frac{1}{4 + i\omega} \right] = i \left[-\frac{i}{(4 + i\omega)^2} \right] = \frac{1}{(4 + i\omega)^2}$$

Fourier transform of basic functions:

Unit impulse function, $f(t) = \delta(t)$

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-i\omega t} dt = \left[e^{-i\omega t} \right]_{t=0} = 1$$

∴ Impulse in time $\Rightarrow$ constant in frequency

Fourier transform of Constant function,
 $f(t) = k$

By Duality property, we have

$$\text{If } \mathcal{F}[f(t)] = F(\omega) \text{ then } \mathcal{F}[F(t)] = 2\pi f(-\omega)$$

We know that $\mathcal{F}[\delta(t)] = 1$

$$\Rightarrow \mathcal{F}[1] = 2\pi\delta(-\omega) = 2\pi\delta(\omega) \quad (\because \text{impulse function is even})$$

$$\therefore \mathcal{F}[k] = 2\pi k\delta(\omega)$$

Hence constant in time $\Rightarrow$ Impulse in frequency

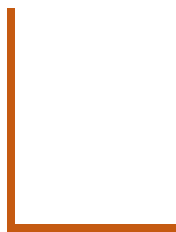
1. $f(t) = e^{iat}$

By frequency shifting property, we have

$$\mathcal{F}[e^{iat}f(t)] = F(\omega - a)$$

$$\text{Also } \mathcal{F}[1] = 2\pi\delta(\omega)$$

$$\therefore \mathcal{F}[1 \cdot e^{iat}] = \mathcal{F}[e^{iat}] = 2\pi\delta(\omega - a)$$



-
1. Trigonometric functions, $f(t) = \cos at$ and $f(t) = \sin at$

$$\mathcal{F}[\cos at] = \mathcal{F}\left[\frac{e^{iat} + e^{-iat}}{2}\right] = \frac{1}{2}\{\mathcal{F}[e^{iat}] + \mathcal{F}[e^{-iat}]\}$$

$$\mathcal{F}[\cos at] = \frac{1}{2}\{2\pi\delta(\omega - a) + 2\pi\delta(\omega + a)\}$$

$$\begin{aligned}\therefore \mathcal{F}[\cos at] &= \pi\delta(\omega - a) + \pi\delta(\omega + a) \\ &= \pi\{\delta(\omega - a) + \delta(\omega + a)\}\end{aligned}$$

$$\mathcal{F}[\sin at] = \mathcal{F}\left[\frac{e^{iat} - e^{-iat}}{2i}\right] = \frac{1}{2i}\{\mathcal{F}[e^{iat}] - \mathcal{F}[e^{-iat}]\}$$

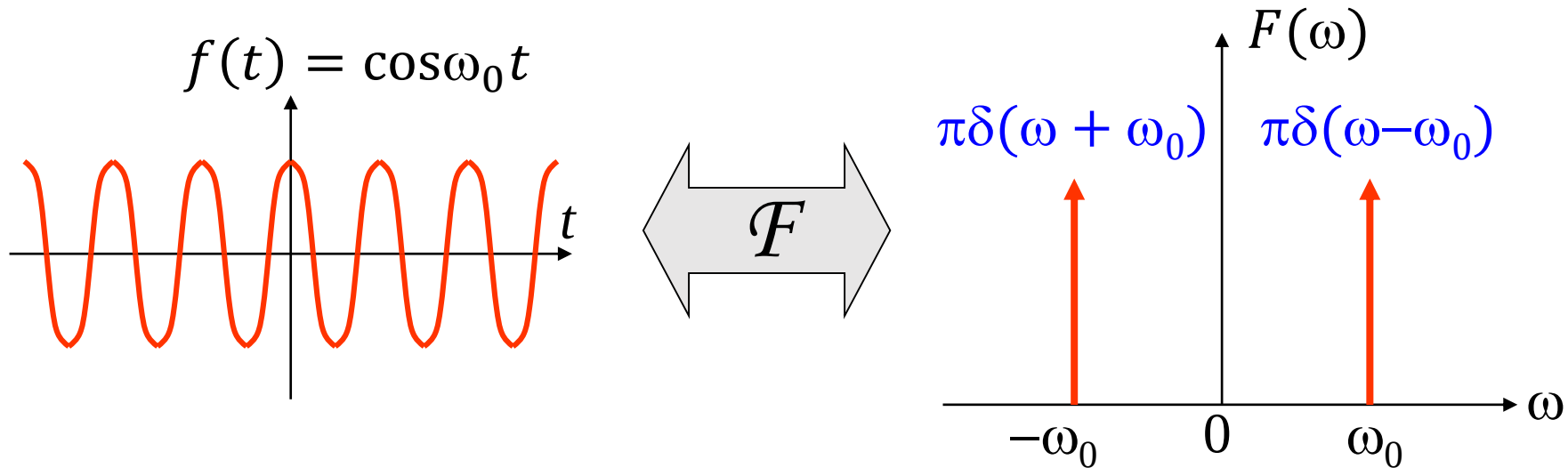
$$\begin{aligned}\mathcal{F}[\sin at] &= \frac{1}{2i}\{2\pi\delta(\omega - a) - 2\pi\delta(\omega + a)\} \\ &= \frac{1}{i}\{\pi\delta(\omega - a) - \pi\delta(\omega + a)\}\end{aligned}$$

$$\therefore \mathcal{F}[\sin at] = -i\pi\delta(\omega - a) + i\pi\delta(\omega + a) = i\pi\{\delta(\omega + a) - \delta(\omega - a)\}$$

Fourier Transforms of Sinusoidal Functions

$$\cos \omega_0 t \xleftrightarrow{\mathcal{F}} \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$$

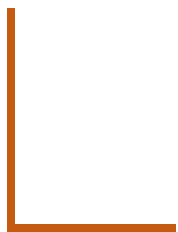
$$\sin(\omega_0 t) \xleftrightarrow{\mathcal{F}} -i\pi \delta(\omega - \omega_0) + i\pi \delta(\omega + \omega_0)$$



- Linearity property:

$$\mathcal{F}[a f(t) + b g(t)] = a \mathcal{F}[f(t)] + b \mathcal{F}[g(t)].$$

- Shifting property: $\mathcal{F}[f(t - a)] = e^{-i\omega a} F(\omega)$
- Frequency shifting property: $\mathcal{F}[e^{iat} f(t)] = F(\omega - a)$
- Modulation Theorem: $\mathcal{F}[f(t) \cos(\omega_0 t)] = \frac{1}{2} [F(\omega + \omega_0) + F(\omega - \omega_0)]$
- Fourier Transform of Derivatives: $\mathcal{F}[f^{(n)}(t)] = (i\omega)^n F(\omega)$,
- Differential with respect to frequency (ω): $\mathcal{F}(t^n f(t)) = i^n F^{(n)}(\omega)$
- Symmetry property $\mathcal{F}[F(t)] = 2\pi f(-\omega)$



-
- $\mathcal{F}[\delta(t)] = 1$
 - $\mathcal{F}[e^{iat}] = 2\pi\delta(\omega - a)$
 - $\mathcal{F}[\cos at] = \pi\{\delta(\omega - a) + \delta(\omega + a)\}$
 - $\mathcal{F}[\sin at] = -i\pi\{\delta(\omega + a) - \delta(\omega - a)\}$
 - $\mathcal{F}[e^{-at}u(t)] = \frac{1}{a+i\omega}$
 - $\mathcal{F}[e^{-a|t|}] = \frac{2a}{a^2+\omega^2}$



Thank you

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